Interval-Width Headroom with Fixed Predictions: A Finite-Sample Audit of Nonlinear Score Information

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Abstract

Can a confidence interval be shortened without acquiring labels, adding unlabeled observations, or changing a predictor? We study binary population means with identical inputs shared across methods. A control-variance decomposition identifies nonlinear conditional-mean information that a scalar power-tuned correction cannot use; the headroom vanishes for affine conditional means within this estimator family. To separate that headroom from finite-sample validity, we construct conservative intervals by exact binomial inversion for both stratum responses and unknown stratum masses, with simultaneous protection for partition selection. We audit 103500 paired trial inputs. In a nearly deterministic nonlinear example at 500 labels, the selection-protected interval is 26.5% narrower than PPI++, with zero observed misses in 3,000 repetitions. This is a conditional result: it is wider on both supplied real-data prediction pools, and selection, sparse strata, and finite unlabeled budgets can erase the gain. Rare-outcome Wald intervals attain only 62.6% coverage at 100 labels, illustrating why shorter is not necessarily better. The contribution is a reproducible validity–width audit, not a new debiasing principle or a universal improvement theorem.

1 Introduction

A scientific analyst may inherit a small labeled sample, a larger unlabeled sample, and an immutable prediction rule. Prediction-powered inference (PPI) uses the labels to correct prediction bias while exploiting the prediction average [1]. Power tuning can protect against an unhelpful scalar prediction [2]. Neither development makes the question of remaining interval-width headroom vacuous: an informative score can have zero linear correlation with the response, finite unlabeled samples introduce uncertainty, and finite-sample coverage need not follow from an asymptotic theorem.

We ask a narrower, falsifiable question: *for a binary population mean, when does nonlinear information in the same fixed score allow shorter intervals, after accounting for estimated score-stratum masses, selection, and coverage?* We select a mean because it directly represents a scientific prevalence, its truth can be known in simulation, and it admits both exact binomial inference and transparent variance algebra. We do not target a realized sample mean or an individual prediction interval. The real-data evaluation below conditions on two finite empirical populations; it is not deployment validation.

The analysis separates three issues often conflated in an efficiency comparison. First, a score may retain information outside its best affine projection. Second, an estimator may learn to exploit that information, yet its Wald interval can under-cover at small sample sizes. Third, a finite-sample interval can retain coverage but lose the apparent width gain to mass estimation and multiple comparisons. We evaluate these issues on exactly the same labeled and unlabeled inputs in each repetition; no label allocation, extra labels, predictor training, or predictor replacement occurs.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

The resulting contributions are deliberately modest. We (i) give an explicit scalar-versus-nonlinear variance identity and its no-headroom condition within a stated control family; (ii) spell out a conservative, implementable finite-sample interval for fixed score partitions with unknown masses and protected partition selection; and (iii) provide a paired audit with negative controls, exact and asymptotic baselines, Monte Carlo uncertainty, and reproducible failures. The statistical ingredients are classical, and stratification already appears in prediction-powered inference [3]. We make no priority claim for score postprocessing, stratification, binomial inversion, or simultaneous selection, and no claim to exhaustive novelty clearance.

2 Setting and existing methods

Let $(Y_i, S_i)_{i=1}^n$ and $(\widetilde{S}_j)_{j=1}^N$ be independent iid samples from the same population, where $Y \in \{0,1\}$ and $S = f(X) \in [0,1]$. The base rule f is fixed independently of these evaluation samples. Our estimand is $\mu = \mathbb{E}Y$. "Fixed data" means that all competing methods receive the same realized inputs within a trial; frequentist coverage is over repetitions of the sampling process. Once all observations and all algorithmic choices are conditioned on, a deterministic interval has no nontrivial repeated-sampling coverage probability left to evaluate.

For a deterministic square-integrable control h, define

$$\widehat{\mu}_h = \overline{Y}_L + \overline{h}_U - \overline{h}_L, \qquad V_h = \frac{\mathrm{Var}(Y - h(S))}{n} + \frac{\mathrm{Var}(h(S))}{N}. \tag{1}$$

This estimator is unbiased for every fixed h, not only accurate predictions. Classical inference uses $h = 0$; the original mean PPI estimator uses $h(S) = S$ [1, Algorithm 1]. The common mean PPI interval is a Wald interval with the two variances in (1) estimated empirically. Its general validity is asymptotic. PPI also gives a finite-sample framework from valid component sets [1, Theorem 1]; finite-sample PPI is not a new idea introduced here.

For scalar power tuning, $h(S) = \lambda S$, PPI++ uses the consistent plug-in

$$\widehat{\lambda} = \frac{\widehat{\mathrm{Cov}}_L(Y, S)}{(1 + n/N)\widehat{\mathrm{Var}}_{L\cup U}(S)}. \tag{2}$$

The denominator pools both score samples, as in Angelopoulos et al. [2, Example 6.1]. We use unrestricted real λ, including negative values: the mean objective remains convex, so clipping to $[0,1]$ is unnecessary. A constant score receives $\widehat{\lambda} = 0$. PPI++'s asymptotic efficiency improvement is not a finite-sample no-worse-width guarantee. We use unbiased empirical variance divisors throughout, rather than the original PPI algorithm's divisors n, N; this does not change first-order limits.

Stratification and scope. Fisch et al. [3, Section 4] study fixed within-stratum sample sizes and stratum weights, with allocation choices and asymptotic confidence intervals. Our labels arrive under iid sampling with random cell counts, and masses must be estimated from the available unlabeled sample. We neither reallocate labels to realize their design nor label our procedure a reproduction of that fixed-allocation algorithm. Fixed-score partition summaries are downstream inference, not changes to the base prediction rule. If a stronger interpretation forbids even downstream score summaries, the nonlinear route is outside that narrower problem. We consider it allowed here while keeping every base score unchanged.

3 What nonlinear information can and cannot buy

Put $r = n/N$ and $m(S) = \mathbb{E}[Y \mid S]$. Write $h_0 = h - \mathbb{E}h$. The following elementary calculation locates the potential gain before any fitting or confidence-interval calibration.

Proposition 1 (Control-family headroom)**.** *Under the assumptions of Section 2,*

$$nV_h = \mathrm{Var}(Y) - \frac{\mathrm{Var}(m(S))}{1+r} + (1+r)\mathbb{E}\left[\left(h_0 - \frac{m(S) - \mu}{1+r}\right)^2\right], \tag{3}$$

$$V_* = \frac{\mathbb{E}\,\mathrm{Var}(Y \mid S)}{n} + \frac{\mathrm{Var}(m(S))}{n+N}. \tag{4}$$

The minimum over fixed score-only controls is attained at $h = m/(1+r) + c$. If $\mathrm{Var}(S) > 0$, the best scalar control has $\lambda_* = \mathrm{Cov}(Y,S)/[(1+r)\,\mathrm{Var}(S)]$ and

$$V_{\mathrm{lin}} - V_* = \frac{\mathrm{Var}(m(S)) - \mathrm{Cov}(Y,S)^2/\,\mathrm{Var}(S)}{n(1+r)} \geq 0. \tag{5}$$

Equality holds exactly when $m(S)$ is affine in S almost surely.

Proof idea. $Y - m(S)$ is orthogonal to every square-integrable score function. Expanding $\mathrm{Var}(Y - h) + r\,\mathrm{Var}(h)$ and completing a square gives (3); minimizing the scalar quadratic gives (5). Cauchy–Schwarz yields nonnegativity and the affine equality condition. Complete proofs are in Appendix A.

The result is *not* a global impossibility theorem for confidence-interval shortening. Affine m leaves no first-order oracle headroom over the optimal scalar within (1), but higher-order calibration or a different estimator class is not ruled out. If m is constant, the optimum is labeled-only variance; if S is constant, every score contrast vanishes. A large gap in (5) is an opportunity, not evidence that a feasible estimated interval will improve.

For a fixed partition $B = B(S)$, let $m_B = \mathbb{E}[Y \mid B]$. Restricting controls to its bin indicators replaces m by m_B, with approximation loss $\mathbb{E}[(m - m_B)^2]/[n(1+r)]$. Refining a partition weakly reduces this oracle loss, but arbitrary increases in bin count do not. A bin-control class also need not contain the scalar score. None of these approximation statements orders finite-sample interval widths.

Our asymptotic nonlinear baseline estimates each bin response mean using *all* labels and sets $\widehat{h}_k = \widehat{p}_k/(1 + n/N)$. With a fixed finite number of independent, prespecified boundaries, positive active cell masses, and $n/N \to \rho < \infty$, its fitted-control remainder is $o_p(n^{-1/2})$. For the residual/unlabeled plug-in variance, $n\widehat{V} \to_p v_\rho$; a Wald theorem requires $v_\rho > 0$. Thus cross-fitting is not required for this fixed-dimensional construction, but sparse strata and zero empirical variance remain small-sample hazards. The proof and degenerate cases appear in Appendix B; no finite-sample unbiasedness is claimed for the fitted map.

4 Finite-sample intervals with unknown stratum masses

A simple exact-in-coverage construction makes the validity cost explicit. Fix a partition $A_1, \ldots, A_K$ independently of both evaluation samples. Let C_k, T_k be its labeled cell count and success count; let D_k be its unlabeled count. Write $p_k = P(Y = 1 \mid S \in A_k)$, $w_k = P(S \in A_k)$, so $\mu = \sum_k w_k p_k$. For positive weights, $T_k \mid (C_1, \ldots, C_K) \sim \mathrm{Bin}(C_k, p_k)$ and $D_k \sim \mathrm{Bin}(N, w_k)$ marginally. This conditions on counts, *not* on every realized score.

For $K \geq 2$, construct equal-tailed Clopper–Pearson (CP) intervals

$$[a_k, b_k] = \mathrm{CP}(T_k, C_k; \alpha/(2K)), \qquad [\ell_k, u_k] = \mathrm{CP}(D_k, N; \alpha/(2K)). \tag{6}$$

Here the third argument is total two-sided error. Empty labeled cells receive $[0,1]$; cells of population mass zero can have arbitrary p_k. Define

$$\mathcal{W} = \{v : \ell_k \leq v_k \leq u_k,\ \sum_k v_k = 1\}, \quad I_B = \left[\min_{v \in \mathcal{W}} \sum_k a_k v_k,\ \max_{v \in \mathcal{W}} \sum_k b_k v_k\right]. \tag{7}$$

For $K = 1$, use labeled-only CP at error α, without wasting an error budget on a known unit mass.

Theorem 1 (Coverage and protected selection)**.** *For every positive n, N and every population satisfying Section 2, $P(\mu \in I_B) \geq 1 - \alpha$. For a prespecified family of M partitions, construct each I_B at error α/M. Selecting the shortest of these corrected intervals, with fixed tie breaking, also has coverage at least $1 - \alpha$.*

Proof idea. Conditional binomial inversion controls each response error even for random counts; binomial inversion also controls each mass error. A union bound covers all $2K$ components. On that event, the true w is feasible in (7), and $a \leq p \leq b$, so the endpoints enclose μ. CP contains each observed proportion, so $D/N \in \mathcal{W}$ and the optimization is always feasible. Another union bound protects arbitrary selection over the prespecified family. No independence among component intervals or among candidates is needed. Appendix A supplies the details.

Both endpoint optimizations are solved by filling the box-constrained simplex in order of the objective coefficients. Our implementation additionally evaluates the corresponding dual bound with outward padding, avoiding reliance on approximate primal feasibility. Standard double-precision beta quantiles are padded outwards; we do not claim formal certification of library special-function error. The theorem is a real-arithmetic statistical guarantee, not a machine-verified floating-point theorem.

The selector, called **Safe CP**, uses four fixed partitions: one bin, score halves, a tails-versus-center split at $1/4, 3/4$, and four equal score bins. Each candidate receives $\alpha/4$. The standalone **Tails CP** uses the tails partition at error α. Protected selection is no wider than its corrected candidates, but need not beat an uncorrected labeled CP or PPI++ interval. Choosing the shortest ordinary 95% candidate ("unsafe selection") has no analogous coverage argument, even if a simulation happens to look conservative.

5 Experiments

5.1 Paired protocol and uncertainty

All intervals have nominal level 95%. Main synthetic experiments use $n \in \{100, 500, 2000\}$, $N = 20n$, and 3,000 independent repetitions per cell, divided into seeds 1729, 2027, 7919. Sensitivity experiments use $n = 500$, $N/n \in \{1, 5, 100\}$, and 1,500 repetitions for affine, sharp-tail, and irrelevant scores. Two fixed real-data prediction pools use the main budgets. There are 39 cells and 103500 distinct paired trial inputs, not that many independent datasets or trained predictors. Fourteen original methods plus two descriptive Hoeffding supplements give 1,656,000 method-output rows. All methods receive identical labeled outcomes and scores and identical unlabeled scores within each repetition. Algorithms never receive unlabeled outcomes.

The fixed synthetic laws use labeled $S \sim U[0, 1]$ and Bernoulli Y with the conditional means in Appendix C. Affine and reversed-affine laws have $m(s) = .05 + .9s$ and $.95 - .9s$. Smooth U uses $.05 + .9(2s - 1)^2$. Sharp tails uses $.005 + .99\mathbf{1}\{s < .25 \text{ or } s \geq .75\}$; its known mean is .5. Irrelevant and constant scores have outcome mean .35. Rare tails has mean .01. A shift stress test changes only unlabeled scores to $\mathrm{Beta}(4, 1)$, violating the common-population assumption. No scenario was dropped after observing outcomes.

Baselines. We compare labeled Wald and CP, original PPI Wald, unrestricted PPI++, full-data fixed-bin Wald controls (half, tails, four and eight bins), standalone fixed-partition CP, Safe CP, unsafe selection, and infeasible known-mass CP. An audit, after mechanics-only smoke checks and declared before full outcome inspection, added labeled Hoeffding and original PPI's union-Hoeffding specialization as descriptive finite-sample baselines; they are not in the locked testing families. An exploratory post-audit mass ablation is separately labeled below. All reported intervals are clipped to $[0, 1]$; zero empirical variance gives a point Wald interval rather than silently removing a repetition.

Coverage error bars are pointwise Wilson 95% Monte Carlo (MC) intervals. Mean-width error bars use 1.96 MC standard errors. Width ratios are ratios of mean widths with a paired delta-method MC interval, not means of per-trial ratios. A declared family of 72 one-sided paired width tests (Safe CP versus PPI++, Tails CP versus labeled CP in 36 nonshift cells) uses Holm correction. Exact binomial undercoverage tests use a separate 252-test Holm family for seven asymptotic methods; exact simultaneous MC lower bounds use a separate 180-member family. These families and their timing are documented in the supplement. Pointwise ratio intervals do not become simultaneous merely because a related width test is adjusted.

5.2 Synthetic headroom and its costs

Table 1, Figure 1, and paired intervals in Table 2 show the intended conditional result. For sharp tails, $\mathrm{Cov}(Y, S) = 0$ but the tails indicator is almost a sufficient response predictor. At 500 labels, Safe CP is 26.5% narrower than PPI++; the pointwise paired MC interval for the reduction is $[26.28, 26.75]\%$. Its observed coverage is 3,000/3,000, not a claim that its true coverage equals one. Standalone Tails CP is 39.8% narrower than labeled CP. At 2,000 labels the Safe CP reduction versus PPI++ is 33.4%. All nine Holm-significant designated width improvements occur in sharp-tail cells; no broad superiority follows.

Table 1: Main synthetic results at $n = 500, N = 10{,}000$. Each entry is mean width / coverage in percent (3,000 repetitions). These are not equal-validity width rankings: CP procedures have finite-sample guarantees under assumptions; Wald procedures are asymptotic. Complete MC uncertainty is in Appendix F.

Scenario	Label CP	PPI++	Bin4 Wald	Tails CP	Safe CP
Affine	0.0893 / 95.6	0.0755 / 95.2	0.0761 / 95.1	0.1615 / 99.9	0.1132 / 99.0
Reversed	0.0893 / 96.1	0.0755 / 94.4	0.0761 / 94.0	0.1614 / 100.0	0.1132 / 99.1
Smooth U	0.0853 / 95.8	0.0835 / 95.0	0.0740 / 94.8	0.1444 / 100.0	0.1081 / 99.1
Sharp tails	0.0893 / 95.8	0.0876 / 95.2	0.0226 / 94.9	0.0538 / 100.0	0.0644 / 100.0
Irrelevant	0.0853 / 95.9	0.0835 / 95.3	0.0833 / 95.1	0.1542 / 100.0	0.1080 / 99.0
Constant	0.0853 / 95.8	0.0836 / 95.2	0.0836 / 95.2	0.1085 / 99.2	0.1080 / 99.0
Rare tails	0.0195 / 97.5	0.0166 / 86.6	0.0165 / 86.6	0.0354 / 99.8	0.0246 / 99.5

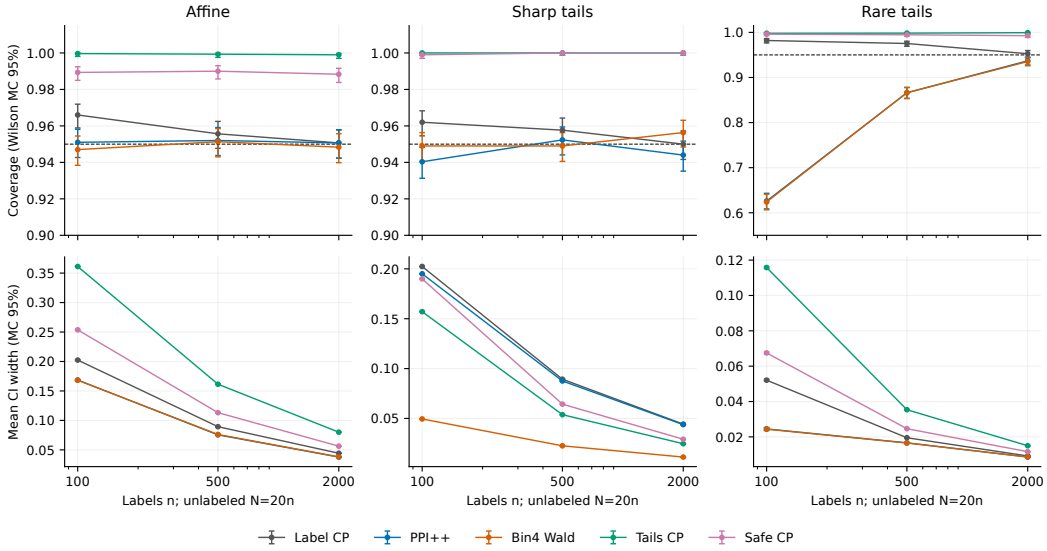


Figure 1: Coverage (top) and width (bottom) on three prespecified laws. Coverage bars are pointwise Wilson MC95; width bars are 1.96 MCSE. Exact construction costs precision on affine data, exploits sharp nonlinear information, and avoids the rare-event Wald collapse. Near-invisible width bars reflect precise MC means, not certainty about future scientific populations.

In smooth U, Bin4 Wald realizes a 11.2% reduction at 2,000 labels, with 94.8% observed coverage, whereas conservative CP gains disappear. On affine and reversed-affine laws, the scalar oracle already captures the relevant conditional mean. Binned Wald controls do not improve first-order efficiency there, and simultaneous exact inference is substantially wider. Irrelevant and constant scores likewise provide no useful headroom. These are retained negative results, not failed cases removed from the comparison.

Finite unlabeled data and selection are not free. At sharp tails and $n = 500$, Safe CP/PPI++ mean-width ratios are $1.292, 1.061, .735, .554$ for $N/n = 1, 5, 20, 100$. The gain reverses at limited unlabeled budgets in this grid; these four points do not estimate a universal crossover. Figure 2 shows this dependence. All same-population exact-family observed coverages range down to 94.3%, from labeled CP; the minimum simultaneous MC lower bound is 92.7%, not 95%. Finite MC frequencies need not exceed nominal coverage in every cell. The three observed exact-method rates below 95% are statistically compatible with nominal after considering their uncertainty; the mathematical proof, not this observation, supplies the finite-sample guarantee.

Table 2: Selected paired ratios with pointwise MC95 intervals. Denominator is labeled CP for the first and fifth rows, PPI++ otherwise. Coverage columns also have pointwise Wilson MC95. Asymptotic Bin4 and finite-sample CP rows have different guarantees; the smooth-U row is descriptive, not a designated width test.

Case	Method	Width ratio	Ratio MC 95%	Coverage	MC 95%
Sharp tails 500	Tails CP	0.602	[0.60005,0.60418]	1.000	[0.999,1.000]
Sharp tails 500	Safe CP	0.735	[0.73255,0.73720]	1.000	[0.999,1.000]
Sharp tails 2000	Safe CP	0.666	[0.66421,0.66738]	1.000	[0.999,1.000]
Smooth U 2000	Bin4 Wald	0.888	[0.88736,0.88803]	0.948	[0.939,0.955]
Irrelevant 500	Safe CP	1.266	[1.26647,1.26647]	0.990	[0.986,0.993]
Galaxies 500	Safe CP	1.873	[1.86994,1.87558]	0.991	[0.987,0.994]

Table 3: Fixed real-prediction empirical populations: mean width / coverage percent. All cells have 3,000 repetitions, $N = 20n$. Repeated source records at large n, N reflect empirical-population sampling, not additional collected data.

Data	n	Label CP	PPI++	Bin4 Wald	Tails CP	Safe CP
Forest	100	0.1485 / 94.4	0.1200 / 92.3	0.1179 / 92.0	0.2488 / 100.0	0.1858 / 99.0
Forest	500	0.0646 / 96.1	0.0535 / 94.5	0.0540 / 94.4	0.1117 / 100.0	0.0817 / 99.3
Forest	2000	0.0319 / 95.4	0.0268 / 95.0	0.0272 / 95.5	0.0554 / 99.9	0.0405 / 98.4
Galaxies	100	0.1785 / 95.3	0.1187 / 93.4	0.1172 / 92.8	0.3034 / 100.0	0.2236 / 99.2
Galaxies	500	0.0785 / 94.9	0.0531 / 94.4	0.0539 / 94.4	0.1372 / 100.0	0.0994 / 99.1
Galaxies	2000	0.0389 / 95.7	0.0265 / 94.3	0.0271 / 94.8	0.0682 / 100.0	0.0494 / 99.3

5.3 Real prediction pools: no exact width win demonstrated

We use the supplied `ppi-python` v0.2.3 forest and galaxies arrays, with 1,596 and 16,743 records. Labels are binary and scores are existing predictions; neither predictor is retrained. Each fixed pool defines an empirical population whose target is its exact label proportion. Labeled and unlabeled observations are drawn independently *with replacement* from this population; source-record overlap does not break independence of independent draws. This identifies repeated-sampling coverage for that empirical law, not the original forest or galaxy superpopulation, and does not create new unique source information. The predictors' original training independence cannot be re-audited from these arrays.

Table 3 and Figure 3 report the negative result. At 500 labels, Safe CP is 52.9% wider than PPI++ for forest and 87.3% wider for galaxies. In both cells it always selects the multiplicity-adjusted labeled candidate. It is also wider than ordinary labeled CP. Nonlinear finite-sample gains in the sharp-tail construction therefore do not transfer to these pools. PPI++ has significant adjusted undercoverage at 100 labels in both datasets, however, so its shorter width is not a universal finite-sample validity result either.

5.4 Failures and ablations

Zero labels and shift expose different collapses. Rare tails at $n = 100$ gives PPI++ coverage 62.6%, versus 98.2% for labeled CP. Its 1,121 intervals $[0, 0]$ account for all but one miss: with zero observed successes the fitted covariance and variance both vanish. The population all-zero probability is $.99^{100}$, so this is a predictable failure mechanism rather than evidence of extra information. There are 31 Holm-significant undercoverage tests across the 252 asymptotic cells/methods, with eight-bin controls particularly fragile at small n. Figure 4 displays the corresponding bin-complexity width comparison; its coverage must still be checked separately.

Under the deliberate score-distribution shift at $n = 500$, PPI++ covers in zero trials; 2,883/3,000 intervals clip to $[1, 1]$. The biased untruncated center moves beyond the support, so its tiny reported width is a clipping pathology, not efficiency. Safe CP happens to select labeled-only in this stress test; that does not extend Theorem 1 to shifted masses. Zero-label and support-clipping point intervals must not be interpreted as the same mechanism.

Isolating mass uncertainty. The original known-mass diagnostic removes mass error and reallocates its error budget to the response intervals, so its gap does *not* isolate the cost of unknown masses. A post-audit exploratory rerun on identical sharp-tail inputs fixes response error at $\alpha/(2K)$ in both variants. At $n = 500, N = 10{,}000$, widths are .053795 (unknown), .029482 (known, matched response budget), and .026207 (known, reallocated). Thus the isolated mass gap is .024313 (paired MCSE .000003), with a separate reallocation gap .003275. This diagnostic uses true weights only as an infeasible ablation; it is not extra information available to any feasible competitor. Complete budget sensitivity is in Appendix D, Table 5.

6 Discussion, limitations, and conclusion

The fixed-input question has a conditional answer. A score may encode strong nonlinear information even when its scalar linear correction is useless; exploiting it can shorten a finite-sample valid interval without new data or a new predictor. Conversely, affine conditional means leave no nonlinear oracle headroom over optimal scalar control in this family. Finite-sample response uncertainty, unknown masses, protected selection, and sparse bins determine whether potential headroom survives in an actual interval.

These conclusions are restricted to binary means, independent common-population sampling, fixed independent partitions, and a small prespecified method family. Our exact construction is conservative, not a shortest-valid-interval algorithm. We have not compared every exact PPI component construction or optimized exact joint multinomial regions. No universal finite-sample dominance or impossibility theorem is established. The unrestricted conditional-mean oracle is not a feasible fitted predictor and is not claimed as a new efficiency bound. The real-data evidence concerns two empirical pools, with no deployment or predictor-training uncertainty assessment. The sharp-tail example is intentionally favorable and nearly deterministic; it is a mechanism test, not evidence of typical scientific effect sizes.

Numerical bounds use standard padded double-precision beta quantiles, not formally verified interval arithmetic. Numerical fallback counts and preclipping centers were not logged in the original runs; endpoint checks cannot prove the absence of all fallback events. Separate method timings are unavailable. The initial implementation and smoke checks, all failed analysis attempts, and the later exploratory ablation are documented rather than presented as a single preregistered study.

Broader impact and research process. Narrow but invalid intervals can create false certainty in scientific and policy decisions; conservative valid intervals can instead make fixed annotation budgets more trustworthy, at a measurable precision cost. The comparisons recommend reporting both coverage and width and disclosing distribution mismatch, rather than selecting whichever interval looks shortest. No human participants, personal records, or high-risk models were collected or released. Original data licenses were not independently resolved; the reproducibility release supplies retrieval instructions and hashes instead of redistributing the raw third-party arrays. This work was conducted by an autonomous language-model research assistant, including design, coding, analysis and drafting, with independent subagent audits; the scripts themselves make no LLM or paid API calls. This disclosure is not a claim of human-author verification.

The practical recommendation is therefore not "always recalibrate." Inspect what information remains in the fixed score, pay explicitly for finite-data and selection uncertainty, and retain the labeled-only alternative. In this audit, genuine exact gains and substantial negative results coexist.

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A Complete proofs

A.1 Binomial inversion and conditional cell counts

For $T \sim \mathrm{Bin}(m, p)$, define the equal-tailed CP interval at error δ by

$$a = \begin{cases} 0 & T = 0, \\ F^{-1}_{\mathrm{Beta}(T, m-T+1)}(\delta/2) & T > 0, \end{cases} \quad b = \begin{cases} 1 & T = m, \\ F^{-1}_{\mathrm{Beta}(T+1, m-T)}(1-\delta/2) & T < m. \end{cases} \tag{8}$$

For $m = 0$ set $[a, b] = [0, 1]$. The lower endpoint inverts the upper binomial tail and the upper endpoint inverts the lower tail. A discrete one-sided tail test at error $\delta/2$ rejects with probability at most $\delta/2$; taking the union of the two rejected tails gives $P_p(p \notin [a, b]) \le \delta$. This standard test-inversion argument also establishes conservatism at boundary probabilities.

For a fixed partition, the joint category probabilities of (B, Y) are $w_k p_k, w_k(1 - p_k)$. Dividing the multinomial joint mass of response and cell counts by the multinomial mass of $C = (C_1, \ldots, C_K)$ yields

$$P(T = t \mid C = c) = \prod_k \binom{c_k}{t_k} p_k^{t_k} (1 - p_k)^{c_k - t_k}. \tag{9}$$

Thus binomial CP covers p_k conditional on each possible count vector, and its unconditional error is also at most its assigned budget by total probability. Empty labeled cells are covered by convention; zero-population-mass cells are empty almost surely. The unlabeled count D_k has marginal binomial law regardless of dependence among different D_k.

Apply these facts with $\delta = \alpha/(2K)$ to the response and mass intervals. Their union failure probability is at most $2K\delta = \alpha$. On the simultaneous coverage event, $w \in \mathcal{W}$ and $a_k \le p_k \le b_k$; because all weights are nonnegative,

$$\min_{v \in \mathcal{W}} a^\top v \le a^\top w \le \mu \le b^\top w \le \max_{v \in \mathcal{W}} b^\top v. \tag{10}$$

The ordinary CP interval contains the empirical proportion for every count (equivalently, the integer binomial mean is a median). Hence $D/N \in \mathcal{W}$, making it nonempty for every observed dataset; it is compact. This proves the fixed-partition claim including empty cells. For $K = 1$ use the binomial argument directly.

For M fixed candidates, each constructed at error α/M, the probability that all cover is at least $1 - \alpha$ by another union bound. On this event any selected candidate covers, even when the selector uses all data. This proves Theorem 1. It does not protect a partition collection first chosen using evaluation data without including those choices in the simultaneous family.

A.2 Box-simplex optimization and dual bounds

For a cost vector c, start with $v = \ell$ and residual $R = 1 - \sum_k \ell_k$. Allocate mass, up to each capacity $u_k - \ell_k$, in ascending order of c_k. Feasibility follows from $\sum \ell \le 1 \le \sum u$. If a feasible vector puts transferable mass on a more expensive coordinate while a cheaper coordinate has unused capacity, transferring ε decreases the objective by $\varepsilon(c_j - c_i)$. Repeated exchanges give the sorted-fill solution, proving the minimum; descending order proves the maximum.

A useful numerical certificate does not require exact primal feasibility. For every real λ and feasible v,

$$c^\top v \ge \lambda + \sum_k (c_k - \lambda) \begin{cases} \ell_k & c_k \ge \lambda, \\ u_k & c_k < \lambda, \end{cases} \tag{11}$$

since $\sum v_k = 1$ and each summand is separately minimized over its box. Replacing minima by maxima gives a universal upper bound. The greedy threshold supplies an optimal λ in exact arithmetic; an approximate threshold still supplies a valid dual bound in exact arithmetic. Code pads component and final endpoints outward by 10^{-10} and fails to $[0, 1]$ if its feasibility/finite checks fail. The number of bins is at most eight, so ordinary objective rounding is far below this pad for inputs in $[0, 1]$. Special-function quantile error has not been formally enclosed, and the statistical theorem is not represented as a formal software proof.

A.3 Proof of Proposition 1

Conditional expectation gives $\mathrm{Cov}(Y, h(S)) = \mathbb{E}[(m(S) - \mu)h_0]$. Independence of the two samples proves (1); expanding its variance gives

$$nV_h = \mathrm{Var}(Y) - 2\mathbb{E}[(m - \mu)h_0] + (1 + r)\mathbb{E}h_0^2. \tag{12}$$

Completing the square proves (3), with equality at $h_0 = (m - \mu)/(1 + r)$. Substituting the law of total variance proves (4). Restricting to $h = \lambda S$ minimizes a scalar quadratic and gives the stated λ_*, with minimum $nV_{\mathrm{lin}} = \mathrm{Var}(Y) - \mathrm{Cov}(Y, S)^2/[(1+r)\,\mathrm{Var}(S)]$. Subtracting (4) gives (5). Since $\mathrm{Cov}(Y, S) = \mathrm{Cov}(m(S), S)$, Cauchy–Schwarz proves nonnegativity; its equality condition for $\mathrm{Var}(S) > 0$ is $m - \mu = c(S - \mathbb{E}S)$ almost surely, including $c = 0$. If S is constant, every score function is constant, and the correction vanishes.

For bin-measurable controls, repeat the argument with $m_B = \mathbb{E}[Y \mid B]$. Orthogonality of conditional expectation gives $\mathrm{Var}(m) - \mathrm{Var}(m_B) = \mathbb{E}[(m - m_B)^2]$, proving the approximation loss. If B' refines B, then $\mathbb{E}[m_{B'} \mid B] = m_B$ and $\mathrm{Var}(m_{B'}) = \mathrm{Var}(m_B) + \mathbb{E}[(m_{B'} - m_B)^2]$. This is the precise refinement ordering, not an ordering by bin count alone. For example, $Y = \mathbf{1}\{S < 1/2\}$ under uniform scores has zero approximation loss with halves but squared approximation error $1/12$ (variance loss $1/[12n(1 + r)]$) with equal thirds. Furthermore the binned indicator span need not contain S, so V_B^* need not dominate V_{lin}.

B Fitted means on fixed bins: assumptions and degeneracies

This result does not learn boundaries. Fix $K < \infty$, discard zero-population-mass cells (which never appear), and assume $w_k > 0$ on each active cell. Let $n, N \to \infty$ with $r_n = n/N \to \rho \in [0, \infty)$. Set $\widehat{p}_k = T_k/C_k$ when $C_k > 0$, with global labeled mean fallback otherwise, and $\widehat{h}_k = \widehat{p}_k/(1 + r_n)$. The corresponding deterministic oracle control is $h_{n,k} = p_k/(1 + r_n)$.

The probability of an empty active cell is at most $\sum_k (1 - w_k)^n \to 0$. For nonempty cells,

$$\widehat{p}_k - p_k = \frac{n^{-1}\sum_i \mathbf{1}\{B_i = k\}(Y_i - p_k)}{C_k/n} = O_p(n^{-1/2}). \tag{13}$$

The bounded fallback on an exponentially rare event does not change this rate. Because K is fixed and r_n bounded,

$$\widehat{\mu}_{\widehat{h}} - \widehat{\mu}_{h_n} = \frac{1}{1 + r_n}\sum_k (\widehat{p}_k - p_k)(D_k/N - C_k/n) = O_p(n^{-1}) = o_p(n^{-1/2}). \tag{14}$$

No independence between the factors in this product is required. Same-sample fitting therefore causes no first-order change; the fitted estimator need not be exactly unbiased.

Write $\sigma_e^2 = \mathbb{E}\,\mathrm{Var}(Y \mid B)$, $\sigma_m^2 = \mathrm{Var}(p_B)$, and $v_\rho = \sigma_e^2 + \rho\sigma_m^2/(1 + \rho)$. Replacing h_n by $h_\rho = p/(1 + \rho)$ changes the estimator by $o_p(n^{-1/2})$, because the coefficient change is $o(1)$ times a bin-frequency contrast. Two independent bounded-sample CLTs then yield

$$\sqrt{n}(\widehat{\mu}_{\widehat{h}} - \mu) \;\Rightarrow\; \mathcal{N}(0, v_\rho). \tag{15}$$

Define the centered, unbiased sample variance estimator

$$\widehat{V} = \frac{s_L^2(Y - \widehat{h}(B))}{n} + \frac{s_U^2(\widehat{h}(B))}{N}. \tag{16}$$

Uniform consistency across the finite collection of bin values, boundedness, and the LLN for fixed-h_ρ moments imply

$$n\widehat{V} = s_L^2(Y - \widehat{h}) + r_n s_U^2(\widehat{h}) \;\to_p\; \mathrm{Var}(Y - h_\rho) + \rho\,\mathrm{Var}(h_\rho) = v_\rho. \tag{17}$$

Unscaled convergence of $\widehat{V}$ to zero alone would not suffice. If $v_\rho > 0$, Slutsky gives asymptotically nominal studentized Wald coverage. There is no uniform assertion over growing K, shrinking cell masses, or population sequences.

If $v_\rho = 0$ with $\rho > 0$, both variance components vanish; Y is constant, so a point interval covers with probability one rather than possessing a nondegenerate normal limit. If $\rho = 0$ and $\sigma_e^2 = 0$, binary

Y is deterministic within each active cell. Once all cells are occupied, fitted values equal the true $p_k \in \{0,1\}$ and the estimator equals

$$\frac{\sum_i Y_i + \sum_j p_{\widetilde{B}_j}}{n+N}. \tag{18}$$

The right side, defined with true p on all datasets, is unconditionally a pooled mean of iid Bernoulli(μ) variables. For $0 < \mu < 1$ it has a $\sqrt{n+N}$ CLT, transferred to the fitted estimator by equality on an event tending to one. On the same event, $(n+N)\widehat{V} \to_p \mu(1-\mu)$; for $\mu \in \{0,1\}$ return to the constant case. One must not condition an iid CLT on occupancy or use only the root-n remainder to assert this faster-rate result. Our experiments have N/n fixed within each cell, and rare-event empirical degeneracy remains a finite-n warning.

For PPI++, consistency of the ideal unrestricted scalar coefficient likewise makes its estimation error times $\overline{S}_U - \overline{S}_L$ negligible at first order. Bounded moments and $\mathrm{Var}(S) > 0$ suffice for this ideal scalar plug-in variance argument. The code additionally zeros coefficients when the pooled score variance is at most 10^{-14}: transferring ideal coefficient consistency to this implementation requires $\mathrm{Var}(S) > 10^{-14}$ under common-population sampling. Below this fixed cutoff the implementation intentionally uses labeled Wald instead; at the threshold no ideal-efficiency guarantee is asserted. None of these statements makes either estimated-control Wald interval finite-sample exact.

C Full experimental specification

C.1 Population laws and partition conventions

Table 4 defines every synthetic generator. Within each nonshift trial, both score samples are independently uniform except in the constant-score law; only labeled outcomes are sampled. All base scores remain immutable. Bernoulli randomization is conditionally independent across labels. The rare-tail law has mean $.001 + .018/2 = .01$; smooth U has mean $.05 + .9/3 = .35$. The half partition

Table 4: Prespecified synthetic populations. Shift is an assumption-violation stress test, not a valid-regime efficiency benchmark.

Scenario	Labeled conditional mean $m(s)$	Truth μ
Affine	$.05 + .9s$	.5
Reverse	$.95 - .9s$	.5
Smooth U	$.05 + .9(2s-1)^2$	.35
Sharp tails	$.005 + .99\mathbf{1}\{s < .25 \text{ or } s \geq .75\}$	.5
Irrelevant	.35, uniform score	.35
Constant	.35, score identically .5	.35
Rare tails	$.001 + .018\mathbf{1}\{s < .25 \text{ or } s \geq .75\}$	.01
Shift	$.05 + .9s$, unlabeled $S \sim \mathrm{Beta}(4,1)$	.5

is $[0, 1/2), [1/2, 1]$. The tails cell is $[0, 1/4) \cup [3/4, 1]$, with center $[1/4, 3/4)$. Equal-score bins are left-closed/right-open except the final bin includes one. The four-bin and eight-bin partitions refine these cells; half and tails are not ordered refinements of one another. Boundaries never use evaluation scores, labels, quantiles, or unlabeled empirical masses.

C.2 Algorithms and finite-sample baseline details

For every Wald interval, use center (1), variance (16), and normal quantile $z_{.975}$. PPI has $h = s$, PPI++ uses (2), fixed-bin controls use $\widehat{p}_k/(1 + n/N)$. Sample variances use $\mathrm{ddof} = 1$. Constant pooled scores get scalar coefficient zero. A small numerical pooled score-variance threshold is 10^{-14}; it is a fixed implementation safeguard, not part of the ideal positive-variance oracle theory. In an empty fitted bin use the global label mean; in an exact response interval use $[0, 1]$. CP endpoints are beta quantiles from (8), with the outward pad described above. Any nonfinite interval calculation detected by the generic Wald/LP checks returns $[0, 1]$; the original run did not log separate fallback counts. All endpoints are componentwise clipped to $[0, 1]$. In valid regimes clipping preserves coverage for a target in this support; under misspecification it can create misleading point intervals.

The labeled Hoeffding baseline uses radius $\sqrt{\log(2/\alpha)/(2n)}$. For original PPI with $S \in [0,1]$, the labeled residual $Y - S \in [-1,1]$ and the unlabeled score range is one. Separate Hoeffding bounds at error $\alpha/2$ give radius

$$2\sqrt{\frac{\log(4/\alpha)}{2n}} + \sqrt{\frac{\log(4/\alpha)}{2N}}. \tag{19}$$

A union bound gives finite-sample coverage under the same-population assumptions. This is a simple existing PPI component-set specialization, not the tightest possible nonasymptotic PPI interval. Both Hoeffding supplements are descriptive; their widths and coverages are included in the complete table.

The original known-mass Tails CP diagnostic uses population weights in place of $\mathcal{W}$ and response budget α/K. It cannot be used as a feasible baseline because weights are not supplied to the real task. The matched exploratory comparison in Appendix D instead holds response budget $\alpha/(2K)$ fixed. Unsafe selection chooses the shortest of ordinary labeled/half/tails/four CP candidates, whereas Safe CP first rebuilds all four candidates at error $\alpha/4$. Neither a true conditional-mean oracle nor true masses are supplied to a feasible competitor.

C.3 MC uncertainty and multiplicity

Each main or empirical seed block contains 1,000 trials; sensitivity blocks contain 500. Trials are the units for MC uncertainty, not the 16 correlated method outputs. For hits H in R trials, the coverage estimate is H/R, with ordinary Wilson95 uncertainty. For width W, MCSE is $s_W/\sqrt{R}$. For paired widths A, B and $\widehat{\theta} = \overline{A}/\overline{B}$, the ratio MCSE is

$$\widehat{\mathrm{SE}}(\widehat{\theta}) = \frac{s_{A-\widehat{\theta}B}}{\sqrt{R}\,\overline{B}}. \tag{20}$$

These ratio intervals are asymptotic pointwise MC intervals. Zero mean denominator is reported undefined, not discarded or set to zero. This occurs for shifted PPI++ at $n = 2000$, where every interval clips to $[1,1]$.

Before full outcome inspection, `multiplicity_manifest.json` fixed: (i) 72 one-sided paired-mean width tests, Safe CP/PPI++ and Tails CP/label CP in all 36 nonshift cells, using Student-t marginal approximations then Holm; (ii) 252 exact undercoverage tests of $H_0 : c \geq .95$, with marginal $P_{.95}\{\mathrm{Bin}(R, .95) \leq H\}$, then a separate Holm correction; (iii) 180 one-sided exact CP MC lower bounds for five feasible exact methods across those 36 cells, with family error .05. For $H > 0$, the latter is $F^{-1}_{\mathrm{Beta}(H,R-H+1)}(.05/180)$, and zero for $H = 0$. Pairing induces dependence across methods, which does not invalidate Holm or the union bound. It also does not make approximate paired t tests exact. The within-declared-family tests are multiplicity-adjusted; the protocol lock is not an externally authenticated preregistration. The documented lock was declared before full outcome inspection, after mechanics-only smoke checks. Smoke results were used only to check mechanics.

The minimum observed exact-family coverage is $2830/3000 = .943333$ for labeled CP under smooth U at $n = 100$, Wilson interval $[.934479, .951054]$ and simultaneous lower bound .927341. Its unadjusted nominal-boundary undercoverage $p = .053263$ does not reject. Exact binomial summation gives that CP procedure's true coverage .954256 in this cell; the unusually low observed rate has probability .003105 under that value, but an exploratory 36-cell Holm check is .111783. These are retained diagnostic calculations from existing outcomes, not new simulation runs, and not evidence that empirical coverage must exceed nominal in every cell. All four non-label exact methods' simultaneous MC lower bounds exceed .95 across this grid; the guarantee remains grounded in the assumptions and proof.

C.4 Data provenance and licensing

Only the forest and galaxies example arrays associated with Angelopoulos et al. [1] are used. They are fetched from the fixed `ppi-python` v0.2.3 loader's Google Drive identifiers; no external predictor is trained and no optional dataset is silently substituted after observing outcomes. The exact targets are $242/1596$ for forest and $4341/16743$ for galaxies. `data_eda.json` records shapes, label prevalence, score ranges, correlations, eight-bin counts, absence of nonfinite values, and SHA256 digests. The arrays contain only outcomes and prediction scores for this study, not personal identifiers.

The original code repository is MIT, but an independent license for these arrays was not established from the provided materials. Therefore the submission source archive excludes those raw arrays and supplies `download_data.py`, source URLs and expected hashes; no new dataset license is asserted.

Our empirical iid-with-replacement design is intentional. Sampling without replacement and pretending observations were independent would instead require a finite-population correction/model. Here exact empirical-population truths are known, independent draws are genuinely independent even with repeated indices, and no claim is made that this resampling represents an independent new scientific validation. Original predictor training independence relative to the original pool is not recoverable from the released arrays; empirical-population resampling conditions on the fixed score/outcome pairs.

C.5 Compute, software, and reproducibility

Experiments used four CPU worker processes, single-thread BLAS/OpenMP, on an Intel Xeon Gold 6230 at 2.10GHz host with 40 visible CPUs; no GPU or paid API was used. Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.6 and matplotlib 3.11.2 were recorded. The primary run took 39.1 seconds wall time and the two descriptive supplements 7.8 seconds; individual cell wall times are in the output JSON files. These are measured host times, not portable performance promises or per-method timings. Peak memory was not monitored in the original run; the working batch size is 40 trials to keep score arrays small, and four-worker memory requirements depend on N. The resource checklist is answered conservatively about missing peak memory.

From the workspace root: install `requirements.txt`; run `python download_data.py`; `python -m pytest -q`; `python run_experiments.py -workers 4`; `python supplement_baselines.py`; `python summarize.py`; `python mass_ablation.py`; and `latexmk -pdf paper.tex`, with the thread limits documented in the README. Use the Python standard-library reproducibility driver for measured stage times and a rerun comparison. Batch size, seed sequence [seed, cell id], cell ordering and NumPy generator are fixed by code. Every empirical main/appendix table except the explicitly labeled analytic-law table is generated by `summarize.py` or `mass_ablation.py`; every numerical figure is generated by `summarize.py`. Full raw endpoints, diagnostics, summaries, and integrity checks are retained. All six initial unit tests passed; they include CP enumeration on a probability grid, independent LP-solver comparisons, boundaries, constant-score behavior, the PPI++ coefficient, protected selection, and an enumerated small stratified sampling law. Numerical tests are not proofs of universal validity.

D Additional evidence and exploratory mass-budget ablation

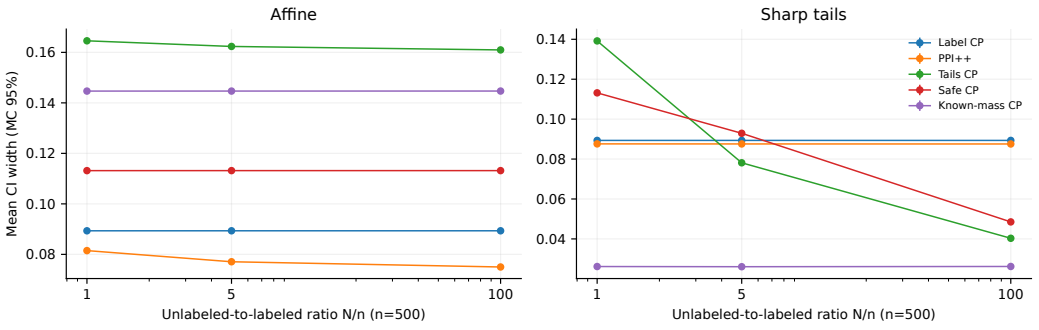


Figure 2: Finite unlabeled-budget sensitivity at 500 labels, with 1.96 MCSE bars. Known-mass CP is infeasible and in this figure also reallocates the response error budget; its gap is not purely the price of estimating masses.

The matched-known procedure covers in every observed ablation repetition, but this does not establish true coverage one. The four mass gaps decrease as N increases while the extra response-budget

Table 5: Exploratory post-audit sharp-tail ablation on the same paired inputs. "Matched" holds response error $\alpha/(2K)$ fixed; "reallocated" uses α/K. Mass gap is unknown minus matched-known width; MCSE uses paired differences. True masses are supplied only to the infeasible diagnostics.

N/n	Unknown weights	Known (matched)	Known (reallocated)	Mass gap	MC SE
1	0.1392	0.0294	0.0262	0.1098	2.0×10^{-5}
5	0.0781	0.0293	0.0261	0.0488	8.3×10^{-6}
20	0.0538	0.0295	0.0262	0.0243	3.0×10^{-6}
100	0.0403	0.0295	0.0262	0.0109	1.9×10^{-6}

reallocation gap stays near .00327. This separates a statistical information effect from a bookkeeping change. The ablation was added after the independent audit noticed the confounding, and is not retroactively included in the locked confirmatory tests.

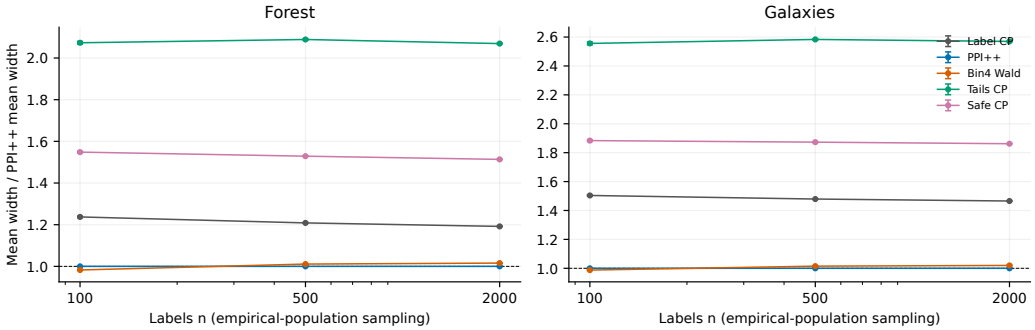


Figure 3: Negative real-pool result: width relative to PPI++ under empirical-population resampling. Bars are paired pointwise MC95 ratio intervals from `paired.csv`; they can be visually small. Marginal width uncertainty and coverage intervals are in Appendix F. This does not certify finite-sample validity of PPI++.

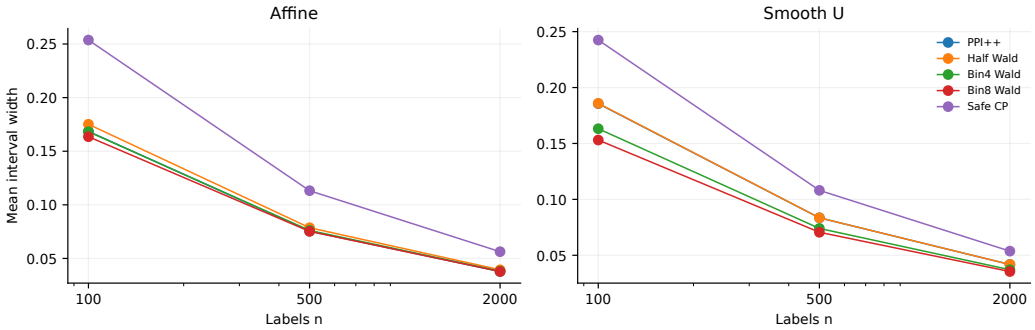


Figure 4: Fixed-bin complexity sensitivity. Refinement helps oracle approximation, not necessarily estimated finite-sample widths or coverage. Eight-bin Wald undercoverage at small sample sizes is retained in the full table.

E Failure log, independence restrictions, and audit scope

The initial source-reading agent could not find `pdftotext`; permitted PDFs downloaded successfully but local extraction failed. A PDF tool then unexpectedly exposed the full permitted PPI++ and Stratified-PPI documents including their bibliographies despite page-limit arguments. Reading stopped as a precaution; no referenced URL was followed and no prohibited identity or confirmed prohibited exposure was recognized. Subsequently `pypdf` physically restricted extraction to permitted main-text pages before inspection. `retrieval_log.md` and `source_verification.md`

document the exact boundaries. Only the three cited user-allowlisted originals were read; no search engines, citation graphs, paper aggregators, or attempts to identify the excluded paper were used. Official conference style, handbook, checklist and ethics pages were consulted solely for formatting/compliance. A separately attempted system resource query was blocked, and CPU information was obtained using `lscpu`; this did not affect experiments.

The first analysis command generated summaries and figures but failed in final EDA because of a missing `counts8` import. Adding the import and rerunning completed it; no observations or numerical method were changed. An initial one-line source extraction command had an unmatched parenthesis and was corrected. Hoeffding/empirical-Bernstein split calibration and trained nonlinear regressors were considered but not implemented or evaluated: the simpler exact binomial route avoids additional fitted-predictor ambiguities and training-label costs. No experimental results are attributed to abandoned methods. The matched-response-budget mass ablation was actually run; it is not a proposed future experiment masquerading as completed evidence.

An independent theory audit rejected unrestricted bin-count monotonicity, required scaled variance consistency and separate degeneracy treatment, and verified the conditional-count/LP/selection construction. An independent analysis audit checked summary-level formulas and selected raw cells; the primary integrity script verified complete pairing, row counts, endpoints, widths and recomputed coverage across all original and supplemental method outputs. The independent audit did not reread every raw output, and neither audit is formal proof-assistant verification or human peer review. Dataset licenses, special-function rounding, numerical-fallback counts, original training independence and deployment generalization remain expressly unresolved rather than silently certified.

F All numerical results

Table 6: All results: mean width ± Monte Carlo SE, coverage and pointwise Wilson 95% Monte Carlo interval. Shift is outside assumptions; unsafe selection and known-mass results are diagnostics.

Cell	Method	Width	Coverage	MC lower	MC upper
Affine 100/2000	Bin4 CP	$0.50188 \pm 3.6 \times 10^{-4}$	1.0000	0.9987	1.0000
Affine 100/2000	Half CP	$0.34451 \pm 2.1 \times 10^{-4}$	1.0000	0.9987	1.0000
Affine 100/2000	Tails CP	$0.36119 \pm 5.8 \times 10^{-5}$	0.9997	0.9981	0.9999
Affine 100/2000	Label CP	$0.20243 \pm 2.4 \times 10^{-5}$	0.9660	0.9589	0.9719
Affine 100/2000	Label Hoeffding	$0.27162 \pm 1.0 \times 10^{-18}$	0.9933	0.9897	0.9957
Affine 100/2000	Label Wald	$0.19602 \pm 2.5 \times 10^{-5}$	0.9460	0.9373	0.9535
Affine 100/2000	Known-mass CP	$0.32449 \pm 6.4 \times 10^{-5}$	0.9993	0.9976	0.9998
Affine 100/2000	PPI	$0.16965 \pm 1.8 \times 10^{-4}$	0.9563	0.9484	0.9631
Affine 100/2000	PPI Hoeffding	0.65828 ± 0	1.0000	0.9987	1.0000
Affine 100/2000	PPI++	$0.16853 \pm 1.6 \times 10^{-4}$	0.9510	0.9427	0.9582
Affine 100/2000	Safe CP	$0.25370 \pm 3.0 \times 10^{-5}$	0.9893	0.9850	0.9924
Affine 100/2000	Unsafe selection	$0.20243 \pm 2.4 \times 10^{-5}$	0.9660	0.9589	0.9719
Affine 100/2000	Bin8 Wald	$0.16354 \pm 1.7 \times 10^{-4}$	0.9347	0.9253	0.9430
Affine 100/2000	Bin4 Wald	$0.16820 \pm 1.6 \times 10^{-4}$	0.9470	0.9384	0.9545
Affine 100/2000	Half Wald	$0.17513 \pm 1.6 \times 10^{-4}$	0.9500	0.9416	0.9572
Affine 100/2000	Tails Wald	$0.19512 \pm 3.5 \times 10^{-5}$	0.9480	0.9395	0.9554
Affine 500/10000	Bin4 CP	$0.23395 \pm 7.6 \times 10^{-5}$	1.0000	0.9987	1.0000
Affine 500/10000	Half CP	$0.15493 \pm 4.0 \times 10^{-5}$	1.0000	0.9987	1.0000
Affine 500/10000	Tails CP	$0.16146 \pm 1.0 \times 10^{-5}$	0.9993	0.9976	0.9998
Affine 500/10000	Label CP	$0.08934 \pm 2.3 \times 10^{-6}$	0.9557	0.9477	0.9625
Affine 500/10000	Label Hoeffding	$0.12147 \pm 5.1 \times 10^{-19}$	0.9940	0.9905	0.9962
Affine 500/10000	Label Wald	$0.08765 \pm 2.3 \times 10^{-6}$	0.9447	0.9359	0.9523
Affine 500/10000	Known-mass CP	$0.14468 \pm 5.9 \times 10^{-6}$	0.9980	0.9956	0.9991
Affine 500/10000	PPI	$0.07588 \pm 3.5 \times 10^{-5}$	0.9530	0.9448	0.9600
Affine 500/10000	PPI Hoeffding	$0.29439 \pm 1.0 \times 10^{-18}$	1.0000	0.9987	1.0000
Affine 500/10000	PPI++	$0.07549 \pm 3.1 \times 10^{-5}$	0.9520	0.9438	0.9591
Affine 500/10000	Safe CP	$0.11317 \pm 2.9 \times 10^{-6}$	0.9900	0.9858	0.9930
Affine 500/10000	Unsafe selection	$0.08934 \pm 2.3 \times 10^{-6}$	0.9557	0.9477	0.9625
Affine 500/10000	Bin8 Wald	$0.07523 \pm 3.2 \times 10^{-5}$	0.9503	0.9420	0.9575
Affine 500/10000	Bin4 Wald	$0.07613 \pm 3.1 \times 10^{-5}$	0.9513	0.9430	0.9585
Affine 500/10000	Half Wald	$0.07864 \pm 3.1 \times 10^{-5}$	0.9493	0.9409	0.9566
Affine 500/10000	Tails Wald	$0.08757 \pm 3.1 \times 10^{-6}$	0.9463	0.9377	0.9538
Affine 2000/40000	Bin4 CP	$0.11678 \pm 1.9 \times 10^{-5}$	1.0000	0.9987	1.0000
Affine 2000/40000	Half CP	$0.07701 \pm 9.9 \times 10^{-6}$	0.9997	0.9981	0.9999
Affine 2000/40000	Tails CP	$0.08004 \pm 2.7 \times 10^{-6}$	0.9990	0.9971	0.9997
Affine 2000/40000	Label CP	$0.04429 \pm 2.9 \times 10^{-7}$	0.9507	0.9423	0.9579
Affine 2000/40000	Label Hoeffding	$0.06074 \pm 8.2 \times 10^{-19}$	0.9930	0.9893	0.9954
Affine 2000/40000	Label Wald	$0.04383 \pm 2.9 \times 10^{-7}$	0.9467	0.9380	0.9542
Affine 2000/40000	Known-mass CP	$0.07175 \pm 7.5 \times 10^{-7}$	0.9970	0.9943	0.9984
Affine 2000/40000	PPI	$0.03797 \pm 8.8 \times 10^{-6}$	0.9523	0.9441	0.9594
Affine 2000/40000	PPI Hoeffding	$0.14720 \pm 5.1 \times 10^{-19}$	1.0000	0.9987	1.0000
Affine 2000/40000	PPI++	$0.03778 \pm 7.7 \times 10^{-6}$	0.9507	0.9423	0.9579
Affine 2000/40000	Safe CP	$0.05629 \pm 3.7 \times 10^{-7}$	0.9883	0.9838	0.9916
Affine 2000/40000	Unsafe selection	$0.04429 \pm 2.9 \times 10^{-7}$	0.9507	0.9423	0.9579
Affine 2000/40000	Bin8 Wald	$0.03782 \pm 7.7 \times 10^{-6}$	0.9493	0.9409	0.9566
Affine 2000/40000	Bin4 Wald	$0.03816 \pm 7.8 \times 10^{-6}$	0.9483	0.9398	0.9557
Affine 2000/40000	Half Wald	$0.03937 \pm 7.6 \times 10^{-6}$	0.9467	0.9380	0.9542
Affine 2000/40000	Tails Wald	$0.04382 \pm 4.0 \times 10^{-7}$	0.9467	0.9380	0.9542
Reversed 100/2000	Bin4 CP	$0.50150 \pm 3.5 \times 10^{-4}$	1.0000	0.9987	1.0000
Reversed 100/2000	Half CP	$0.34403 \pm 2.0 \times 10^{-4}$	0.9997	0.9981	0.9999
Reversed 100/2000	Tails CP	$0.36128 \pm 6.0 \times 10^{-5}$	1.0000	0.9987	1.0000
Reversed 100/2000	Label CP	$0.20245 \pm 2.4 \times 10^{-5}$	0.9697	0.9629	0.9752
Reversed 100/2000	Label Hoeffding	$0.27162 \pm 1.0 \times 10^{-18}$	0.9930	0.9893	0.9954

Cell	Method	Width	Coverage	MC lower	MC upper
Reversed 100/2000	Label Wald	$0.19604 \pm 2.5 \times 10^{-5}$	0.9530	0.9448	0.9600
Reversed 100/2000	Known-mass CP	$0.32445 \pm 6.4 \times 10^{-5}$	0.9993	0.9976	0.9998
Reversed 100/2000	PPI	$0.27395 \pm 1.6 \times 10^{-4}$	0.9523	0.9441	0.9594
Reversed 100/2000	PPI Hoeffding	$0.65803 \pm 5.1 \times 10^{-5}$	1.0000	0.9987	1.0000
Reversed 100/2000	PPI++	$0.16816 \pm 1.5 \times 10^{-4}$	0.9493	0.9409	0.9566
Reversed 100/2000	Safe CP	$0.25373 \pm 3.0 \times 10^{-5}$	0.9883	0.9838	0.9916
Reversed 100/2000	Unsafe selection	$0.20245 \pm 2.4 \times 10^{-5}$	0.9697	0.9629	0.9752
Reversed 100/2000	Bin8 Wald	$0.16329 \pm 1.6 \times 10^{-4}$	0.9343	0.9249	0.9427
Reversed 100/2000	Bin4 Wald	$0.16786 \pm 1.6 \times 10^{-4}$	0.9427	0.9338	0.9504
Reversed 100/2000	Half Wald	$0.17473 \pm 1.5 \times 10^{-4}$	0.9470	0.9384	0.9545
Reversed 100/2000	Tails Wald	$0.19509 \pm 3.5 \times 10^{-5}$	0.9523	0.9441	0.9594
Reversed 500/10000	Bin4 CP	$0.23383 \pm 7.7 \times 10^{-5}$	1.0000	0.9987	1.0000
Reversed 500/10000	Half CP	$0.15498 \pm 4.0 \times 10^{-5}$	0.9997	0.9981	0.9999
Reversed 500/10000	Tails CP	$0.16145 \pm 9.8 \times 10^{-6}$	0.9997	0.9981	0.9999
Reversed 500/10000	Label CP	$0.08934 \pm 2.2 \times 10^{-6}$	0.9607	0.9531	0.9671
Reversed 500/10000	Label Hoeffding	$0.12147 \pm 5.1 \times 10^{-19}$	0.9957	0.9926	0.9975
Reversed 500/10000	Label Wald	$0.08765 \pm 2.2 \times 10^{-6}$	0.9517	0.9434	0.9588
Reversed 500/10000	Known-mass CP	$0.14469 \pm 5.7 \times 10^{-6}$	0.9987	0.9966	0.9995
Reversed 500/10000	PPI	$0.12238 \pm 3.0 \times 10^{-5}$	0.9580	0.9502	0.9646
Reversed 500/10000	PPI Hoeffding	$0.29439 \pm 1.0 \times 10^{-18}$	1.0000	0.9987	1.0000
Reversed 500/10000	PPI++	$0.07550 \pm 3.1 \times 10^{-5}$	0.9437	0.9348	0.9514
Reversed 500/10000	Safe CP	$0.11317 \pm 2.8 \times 10^{-6}$	0.9913	0.9873	0.9941
Reversed 500/10000	Unsafe selection	$0.08934 \pm 2.2 \times 10^{-6}$	0.9607	0.9531	0.9671
Reversed 500/10000	Bin8 Wald	$0.07524 \pm 3.1 \times 10^{-5}$	0.9410	0.9320	0.9489
Reversed 500/10000	Bin4 Wald	$0.07613 \pm 3.1 \times 10^{-5}$	0.9403	0.9313	0.9483
Reversed 500/10000	Half Wald	$0.07868 \pm 3.0 \times 10^{-5}$	0.9433	0.9345	0.9511
Reversed 500/10000	Tails Wald	$0.08757 \pm 3.0 \times 10^{-6}$	0.9530	0.9448	0.9600
Reversed 2000/40000	Bin4 CP	$0.11675 \pm 1.9 \times 10^{-5}$	1.0000	0.9987	1.0000
Reversed 2000/40000	Half CP	$0.07699 \pm 1.0 \times 10^{-5}$	1.0000	0.9987	1.0000
Reversed 2000/40000	Tails CP	$0.08004 \pm 2.7 \times 10^{-6}$	0.9993	0.9976	0.9998
Reversed 2000/40000	Label CP	$0.04429 \pm 2.8 \times 10^{-7}$	0.9513	0.9430	0.9585
Reversed 2000/40000	Label Hoeffding	$0.06074 \pm 7.7 \times 10^{-19}$	0.9940	0.9905	0.9962
Reversed 2000/40000	Label Wald	$0.04383 \pm 2.8 \times 10^{-7}$	0.9453	0.9366	0.9529
Reversed 2000/40000	Known-mass CP	$0.07175 \pm 7.6 \times 10^{-7}$	0.9987	0.9966	0.9995
Reversed 2000/40000	PPI	$0.06120 \pm 7.5 \times 10^{-6}$	0.9477	0.9391	0.9551
Reversed 2000/40000	PPI Hoeffding	$0.14720 \pm 5.1 \times 10^{-19}$	1.0000	0.9987	1.0000
Reversed 2000/40000	PPI++	$0.03776 \pm 7.8 \times 10^{-6}$	0.9473	0.9388	0.9548
Reversed 2000/40000	Safe CP	$0.05628 \pm 3.6 \times 10^{-7}$	0.9903	0.9862	0.9933
Reversed 2000/40000	Unsafe selection	$0.04429 \pm 2.8 \times 10^{-7}$	0.9513	0.9430	0.9585
Reversed 2000/40000	Bin8 Wald	$0.03781 \pm 7.8 \times 10^{-6}$	0.9457	0.9370	0.9532
Reversed 2000/40000	Bin4 Wald	$0.03815 \pm 7.9 \times 10^{-6}$	0.9510	0.9427	0.9582
Reversed 2000/40000	Half Wald	$0.03936 \pm 7.7 \times 10^{-6}$	0.9490	0.9405	0.9563
Reversed 2000/40000	Tails Wald	$0.04382 \pm 4.1 \times 10^{-7}$	0.9470	0.9384	0.9545
Smooth U 100/2000	Bin4 CP	$0.48788 \pm 3.9 \times 10^{-4}$	1.0000	0.9987	1.0000
Smooth U 100/2000	Half CP	$0.34564 \pm 2.0 \times 10^{-4}$	1.0000	0.9987	1.0000
Smooth U 100/2000	Tails CP	$0.32202 \pm 3.0 \times 10^{-4}$	1.0000	0.9987	1.0000
Smooth U 100/2000	Label CP	$0.19352 \pm 1.1 \times 10^{-4}$	0.9433	0.9345	0.9511
Smooth U 100/2000	Label Hoeffding	$0.27162 \pm 1.0 \times 10^{-18}$	0.9943	0.9909	0.9965
Smooth U 100/2000	Label Wald	$0.18684 \pm 1.2 \times 10^{-4}$	0.9283	0.9185	0.9370
Smooth U 100/2000	Known-mass CP	$0.27163 \pm 3.1 \times 10^{-4}$	0.9993	0.9976	0.9998
Smooth U 100/2000	PPI	$0.21920 \pm 2.5 \times 10^{-4}$	0.9377	0.9284	0.9458
Smooth U 100/2000	PPI Hoeffding	$0.64389 \pm 4.8 \times 10^{-4}$	1.0000	0.9987	1.0000
Smooth U 100/2000	PPI++	$0.18560 \pm 1.2 \times 10^{-4}$	0.9277	0.9178	0.9364
Smooth U 100/2000	Safe CP	$0.24248 \pm 1.4 \times 10^{-4}$	0.9877	0.9830	0.9910
Smooth U 100/2000	Unsafe selection	$0.19352 \pm 1.1 \times 10^{-4}$	0.9433	0.9345	0.9511
Smooth U 100/2000	Bin8 Wald	$0.15308 \pm 1.9 \times 10^{-4}$	0.9193	0.9090	0.9286
Smooth U 100/2000	Bin4 Wald	$0.16318 \pm 1.7 \times 10^{-4}$	0.9297	0.9200	0.9383
Smooth U 100/2000	Half Wald	$0.18589 \pm 1.2 \times 10^{-4}$	0.9300	0.9203	0.9386
Smooth U 100/2000	Tails Wald	$0.16478 \pm 1.7 \times 10^{-4}$	0.9343	0.9249	0.9427

Cell	Method	Width	Coverage	MC lower	MC upper
Smooth U 500/10000	Bin4 CP	$0.22649 \pm 9.0 \times 10^{-5}$	1.0000	0.9987	1.0000
Smooth U 500/10000	Half CP	$0.15425 \pm 3.9 \times 10^{-5}$	1.0000	0.9987	1.0000
Smooth U 500/10000	Tails CP	$0.14438 \pm 6.0 \times 10^{-5}$	0.9997	0.9981	0.9999
Smooth U 500/10000	Label CP	$0.08533 \pm 2.1 \times 10^{-5}$	0.9583	0.9506	0.9649
Smooth U 500/10000	Label Hoeffding	$0.12147 \pm 5.1 \times 10^{-19}$	0.9967	0.9939	0.9982
Smooth U 500/10000	Label Wald	$0.08362 \pm 2.2 \times 10^{-5}$	0.9517	0.9434	0.9588
Smooth U 500/10000	Known-mass CP	$0.12016 \pm 6.4 \times 10^{-5}$	0.9990	0.9971	0.9997
Smooth U 500/10000	PPI	$0.09835 \pm 4.8 \times 10^{-5}$	0.9457	0.9370	0.9532
Smooth U 500/10000	PPI Hoeffding	$0.29439 \pm 1.0 \times 10^{-18}$	1.0000	0.9987	1.0000
Smooth U 500/10000	PPI++	$0.08351 \pm 2.2 \times 10^{-5}$	0.9497	0.9413	0.9569
Smooth U 500/10000	Safe CP	$0.10807 \pm 2.7 \times 10^{-5}$	0.9910	0.9869	0.9938
Smooth U 500/10000	Unsafe selection	$0.08533 \pm 2.1 \times 10^{-5}$	0.9583	0.9506	0.9649
Smooth U 500/10000	Bin8 Wald	$0.07059 \pm 3.6 \times 10^{-5}$	0.9493	0.9409	0.9566
Smooth U 500/10000	Bin4 Wald	$0.07402 \pm 3.2 \times 10^{-5}$	0.9483	0.9398	0.9557
Smooth U 500/10000	Half Wald	$0.08353 \pm 2.2 \times 10^{-5}$	0.9510	0.9427	0.9582
Smooth U 500/10000	Tails Wald	$0.07416 \pm 3.2 \times 10^{-5}$	0.9477	0.9391	0.9551
Smooth U 2000/40000	Bin4 CP	$0.11300 \pm 2.2 \times 10^{-5}$	1.0000	0.9987	1.0000
Smooth U 2000/40000	Half CP	$0.07638 \pm 1.0 \times 10^{-5}$	1.0000	0.9987	1.0000
Smooth U 2000/40000	Tails CP	$0.07166 \pm 1.5 \times 10^{-5}$	1.0000	0.9987	1.0000
Smooth U 2000/40000	Label CP	$0.04226 \pm 5.4 \times 10^{-6}$	0.9527	0.9445	0.9597
Smooth U 2000/40000	Label Hoeffding	$0.06074 \pm 2.5 \times 10^{-19}$	0.9947	0.9914	0.9967
Smooth U 2000/40000	Label Wald	$0.04180 \pm 5.4 \times 10^{-6}$	0.9487	0.9402	0.9560
Smooth U 2000/40000	Known-mass CP	$0.05941 \pm 1.5 \times 10^{-5}$	1.0000	0.9987	1.0000
Smooth U 2000/40000	PPI	$0.04919 \pm 1.2 \times 10^{-5}$	0.9560	0.9481	0.9628
Smooth U 2000/40000	PPI Hoeffding	$0.14720 \pm 5.1 \times 10^{-19}$	1.0000	0.9987	1.0000
Smooth U 2000/40000	PPI++	$0.04178 \pm 5.4 \times 10^{-6}$	0.9477	0.9391	0.9551
Smooth U 2000/40000	Safe CP	$0.05371 \pm 6.9 \times 10^{-6}$	0.9900	0.9858	0.9930
Smooth U 2000/40000	Unsafe selection	$0.04226 \pm 5.4 \times 10^{-6}$	0.9527	0.9445	0.9597
Smooth U 2000/40000	Bin8 Wald	$0.03547 \pm 9.1 \times 10^{-6}$	0.9477	0.9391	0.9551
Smooth U 2000/40000	Bin4 Wald	$0.03709 \pm 7.9 \times 10^{-6}$	0.9480	0.9395	0.9554
Smooth U 2000/40000	Half Wald	$0.04179 \pm 5.4 \times 10^{-6}$	0.9483	0.9398	0.9557
Smooth U 2000/40000	Tails Wald	$0.03711 \pm 7.9 \times 10^{-6}$	0.9480	0.9395	0.9554
Sharp tails 100/2000	Bin4 CP	$0.30113 \pm 2.2 \times 10^{-4}$	1.0000	0.9987	1.0000
Sharp tails 100/2000	Half CP	$0.36108 \pm 6.3 \times 10^{-5}$	0.9997	0.9981	0.9999
Sharp tails 100/2000	Tails CP	$0.15707 \pm 2.3 \times 10^{-4}$	1.0000	0.9987	1.0000
Sharp tails 100/2000	Label CP	$0.20237 \pm 2.6 \times 10^{-5}$	0.9620	0.9545	0.9683
Sharp tails 100/2000	Label Hoeffding	$0.27162 \pm 1.0 \times 10^{-18}$	0.9913	0.9873	0.9941
Sharp tails 100/2000	Label Wald	$0.19597 \pm 2.7 \times 10^{-5}$	0.9413	0.9323	0.9492
Sharp tails 100/2000	Known-mass CP	$0.09370 \pm 2.3 \times 10^{-4}$	1.0000	0.9987	1.0000
Sharp tails 100/2000	PPI	$0.22756 \pm 1.8 \times 10^{-4}$	0.9437	0.9348	0.9514
Sharp tails 100/2000	PPI Hoeffding	$0.65822 \pm 3.0 \times 10^{-5}$	1.0000	0.9987	1.0000
Sharp tails 100/2000	PPI++	$0.19505 \pm 3.6 \times 10^{-5}$	0.9403	0.9313	0.9483
Sharp tails 100/2000	Safe CP	$0.18995 \pm 2.4 \times 10^{-4}$	0.9990	0.9971	0.9997
Sharp tails 100/2000	Unsafe selection	$0.15702 \pm 2.2 \times 10^{-4}$	0.9977	0.9952	0.9989
Sharp tails 100/2000	Bin8 Wald	$0.04926 \pm 1.6 \times 10^{-4}$	0.9473	0.9388	0.9548
Sharp tails 100/2000	Bin4 Wald	$0.04950 \pm 1.7 \times 10^{-4}$	0.9490	0.9405	0.9563
Sharp tails 100/2000	Half Wald	$0.19506 \pm 3.5 \times 10^{-5}$	0.9407	0.9316	0.9486
Sharp tails 100/2000	Tails Wald	$0.04962 \pm 1.7 \times 10^{-4}$	0.9497	0.9413	0.9569
Sharp tails 500/10000	Bin4 CP	$0.09987 \pm 1.1 \times 10^{-4}$	1.0000	0.9987	1.0000
Sharp tails 500/10000	Half CP	$0.16147 \pm 9.9 \times 10^{-6}$	0.9997	0.9981	0.9999
Sharp tails 500/10000	Tails CP	$0.05379 \pm 9.4 \times 10^{-5}$	1.0000	0.9987	1.0000
Sharp tails 500/10000	Label CP	$0.08934 \pm 2.3 \times 10^{-6}$	0.9577	0.9499	0.9643
Sharp tails 500/10000	Label Hoeffding	$0.12147 \pm 5.1 \times 10^{-19}$	0.9930	0.9893	0.9954
Sharp tails 500/10000	Label Wald	$0.08765 \pm 2.3 \times 10^{-6}$	0.9490	0.9405	0.9563
Sharp tails 500/10000	Known-mass CP	$0.02621 \pm 9.1 \times 10^{-5}$	1.0000	0.9987	1.0000
Sharp tails 500/10000	PPI	$0.10184 \pm 3.7 \times 10^{-5}$	0.9527	0.9445	0.9597
Sharp tails 500/10000	PPI Hoeffding	$0.29439 \pm 1.0 \times 10^{-18}$	1.0000	0.9987	1.0000
Sharp tails 500/10000	PPI++	$0.08757 \pm 3.1 \times 10^{-6}$	0.9523	0.9441	0.9594
Sharp tails 500/10000	Safe CP	$0.06435 \pm 1.0 \times 10^{-4}$	1.0000	0.9987	1.0000

Cell	Method	Width	Coverage	MC lower	MC upper
Sharp tails 500/10000	Unsafe selection	$0.05379 \pm 9.4 \times 10^{-5}$	1.0000	0.9987	1.0000
Sharp tails 500/10000	Bin8 Wald	$0.02254 \pm 3.6 \times 10^{-5}$	0.9487	0.9402	0.9560
Sharp tails 500/10000	Bin4 Wald	$0.02256 \pm 3.7 \times 10^{-5}$	0.9490	0.9405	0.9563
Sharp tails 500/10000	Half Wald	$0.08757 \pm 3.2 \times 10^{-6}$	0.9513	0.9430	0.9585
Sharp tails 500/10000	Tails Wald	$0.02258 \pm 3.7 \times 10^{-5}$	0.9510	0.9427	0.9582
Sharp tails 2000/40000	Bin4 CP	$0.04349 \pm 4.2 \times 10^{-5}$	1.0000	0.9987	1.0000
Sharp tails 2000/40000	Half CP	$0.08004 \pm 2.7 \times 10^{-6}$	0.9990	0.9971	0.9997
Sharp tails 2000/40000	Tails CP	$0.02467 \pm 3.1 \times 10^{-5}$	1.0000	0.9987	1.0000
Sharp tails 2000/40000	Label CP	$0.04429 \pm 2.9 \times 10^{-7}$	0.9500	0.9416	0.9572
Sharp tails 2000/40000	Label Hoeffding	$0.06074 \pm 7.9 \times 10^{-19}$	0.9933	0.9897	0.9957
Sharp tails 2000/40000	Label Wald	$0.04383 \pm 3.0 \times 10^{-7}$	0.9433	0.9345	0.9511
Sharp tails 2000/40000	Known-mass CP	$0.01110 \pm 2.8 \times 10^{-5}$	1.0000	0.9987	1.0000
Sharp tails 2000/40000	PPI	$0.05092 \pm 9.3 \times 10^{-6}$	0.9497	0.9413	0.9569
Sharp tails 2000/40000	PPI Hoeffding	$0.14720 \pm 5.1 \times 10^{-19}$	1.0000	0.9987	1.0000
Sharp tails 2000/40000	PPI++	$0.04382 \pm 4.1 \times 10^{-7}$	0.9440	0.9352	0.9517
Sharp tails 2000/40000	Safe CP	$0.02917 \pm 3.5 \times 10^{-5}$	1.0000	0.9987	1.0000
Sharp tails 2000/40000	Unsafe selection	$0.02467 \pm 3.1 \times 10^{-5}$	1.0000	0.9987	1.0000
Sharp tails 2000/40000	Bin8 Wald	$0.01128 \pm 9.2 \times 10^{-6}$	0.9553	0.9473	0.9622
Sharp tails 2000/40000	Bin4 Wald	$0.01128 \pm 9.2 \times 10^{-6}$	0.9563	0.9484	0.9631
Sharp tails 2000/40000	Half Wald	$0.04382 \pm 4.0 \times 10^{-7}$	0.9460	0.9373	0.9535
Sharp tails 2000/40000	Tails Wald	$0.01129 \pm 9.2 \times 10^{-6}$	0.9563	0.9484	0.9631
Irrelevant 100/2000	Bin4 CP	$0.52007 \pm 3.0 \times 10^{-4}$	1.0000	0.9987	1.0000
Irrelevant 100/2000	Half CP	$0.34563 \pm 2.0 \times 10^{-4}$	0.9997	0.9981	0.9999
Irrelevant 100/2000	Tails CP	$0.34565 \pm 2.0 \times 10^{-4}$	1.0000	0.9987	1.0000
Irrelevant 100/2000	Label CP	$0.19354 \pm 1.1 \times 10^{-4}$	0.9557	0.9477	0.9625
Irrelevant 100/2000	Label Hoeffding	$0.27162 \pm 1.0 \times 10^{-18}$	0.9937	0.9901	0.9959
Irrelevant 100/2000	Label Wald	$0.18686 \pm 1.1 \times 10^{-4}$	0.9443	0.9355	0.9520
Irrelevant 100/2000	Known-mass CP	$0.31041 \pm 1.8 \times 10^{-4}$	0.9983	0.9961	0.9993
Irrelevant 100/2000	PPI	$0.21916 \pm 2.1 \times 10^{-4}$	0.9507	0.9423	0.9579
Irrelevant 100/2000	PPI Hoeffding	$0.64511 \pm 4.5 \times 10^{-4}$	1.0000	0.9987	1.0000
Irrelevant 100/2000	PPI++	$0.18595 \pm 1.1 \times 10^{-4}$	0.9440	0.9352	0.9517
Irrelevant 100/2000	Safe CP	$0.24251 \pm 1.4 \times 10^{-4}$	0.9890	0.9846	0.9922
Irrelevant 100/2000	Unsafe selection	$0.19354 \pm 1.1 \times 10^{-4}$	0.9557	0.9477	0.9625
Irrelevant 100/2000	Bin8 Wald	$0.18039 \pm 1.3 \times 10^{-4}$	0.9313	0.9217	0.9398
Irrelevant 100/2000	Bin4 Wald	$0.18407 \pm 1.2 \times 10^{-4}$	0.9390	0.9299	0.9470
Irrelevant 100/2000	Half Wald	$0.18594 \pm 1.1 \times 10^{-4}$	0.9447	0.9359	0.9523
Irrelevant 100/2000	Tails Wald	$0.18593 \pm 1.1 \times 10^{-4}$	0.9443	0.9355	0.9520
Irrelevant 500/10000	Bin4 CP	$0.23925 \pm 6.1 \times 10^{-5}$	1.0000	0.9987	1.0000
Irrelevant 500/10000	Half CP	$0.15416 \pm 4.0 \times 10^{-5}$	0.9997	0.9981	0.9999
Irrelevant 500/10000	Tails CP	$0.15416 \pm 4.0 \times 10^{-5}$	0.9997	0.9981	0.9999
Irrelevant 500/10000	Label CP	$0.08528 \pm 2.2 \times 10^{-5}$	0.9590	0.9513	0.9655
Irrelevant 500/10000	Label Hoeffding	$0.12147 \pm 5.1 \times 10^{-19}$	0.9967	0.9939	0.9982
Irrelevant 500/10000	Label Wald	$0.08357 \pm 2.2 \times 10^{-5}$	0.9530	0.9448	0.9600
Irrelevant 500/10000	Known-mass CP	$0.13814 \pm 3.5 \times 10^{-5}$	0.9983	0.9961	0.9993
Irrelevant 500/10000	PPI	$0.09831 \pm 4.1 \times 10^{-5}$	0.9493	0.9409	0.9566
Irrelevant 500/10000	PPI Hoeffding	$0.29439 \pm 1.0 \times 10^{-18}$	1.0000	0.9987	1.0000
Irrelevant 500/10000	PPI++	$0.08348 \pm 2.2 \times 10^{-5}$	0.9527	0.9445	0.9597
Irrelevant 500/10000	Safe CP	$0.10800 \pm 2.7 \times 10^{-5}$	0.9900	0.9858	0.9930
Irrelevant 500/10000	Unsafe selection	$0.08528 \pm 2.2 \times 10^{-5}$	0.9590	0.9513	0.9655
Irrelevant 500/10000	Bin8 Wald	$0.08301 \pm 2.2 \times 10^{-5}$	0.9483	0.9398	0.9557
Irrelevant 500/10000	Bin4 Wald	$0.08332 \pm 2.2 \times 10^{-5}$	0.9510	0.9427	0.9582
Irrelevant 500/10000	Half Wald	$0.08348 \pm 2.2 \times 10^{-5}$	0.9527	0.9445	0.9597
Irrelevant 500/10000	Tails Wald	$0.08348 \pm 2.2 \times 10^{-5}$	0.9517	0.9434	0.9588
Irrelevant 2000/40000	Bin4 CP	$0.11875 \pm 1.6 \times 10^{-5}$	1.0000	0.9987	1.0000
Irrelevant 2000/40000	Half CP	$0.07639 \pm 1.0 \times 10^{-5}$	0.9997	0.9981	0.9999
Irrelevant 2000/40000	Tails CP	$0.07639 \pm 1.0 \times 10^{-5}$	0.9997	0.9981	0.9999
Irrelevant 2000/40000	Label CP	$0.04227 \pm 5.4 \times 10^{-6}$	0.9560	0.9481	0.9628
Irrelevant 2000/40000	Label Hoeffding	$0.06074 \pm 2.5 \times 10^{-19}$	0.9960	0.9930	0.9977
Irrelevant 2000/40000	Label Wald	$0.04180 \pm 5.4 \times 10^{-6}$	0.9523	0.9441	0.9594

Cell	Method	Width	Coverage	MC lower	MC upper
Irrelevant 2000/40000	Known-mass CP	$0.06848 \pm 8.7 \times 10^{-6}$	0.9987	0.9966	0.9995
Irrelevant 2000/40000	PPI	$0.04921 \pm 1.0 \times 10^{-5}$	0.9523	0.9441	0.9594
Irrelevant 2000/40000	PPI Hoeffding	$0.14720 \pm 5.1 \times 10^{-19}$	1.0000	0.9987	1.0000
Irrelevant 2000/40000	PPI++	$0.04179 \pm 5.4 \times 10^{-6}$	0.9500	0.9416	0.9572
Irrelevant 2000/40000	Safe CP	$0.05371 \pm 6.9 \times 10^{-6}$	0.9870	0.9823	0.9905
Irrelevant 2000/40000	Unsafe selection	$0.04227 \pm 5.4 \times 10^{-6}$	0.9560	0.9481	0.9628
Irrelevant 2000/40000	Bin8 Wald	$0.04173 \pm 5.4 \times 10^{-6}$	0.9497	0.9413	0.9569
Irrelevant 2000/40000	Bin4 Wald	$0.04177 \pm 5.4 \times 10^{-6}$	0.9490	0.9405	0.9563
Irrelevant 2000/40000	Half Wald	$0.04179 \pm 5.4 \times 10^{-6}$	0.9497	0.9413	0.9569
Irrelevant 2000/40000	Tails Wald	$0.04179 \pm 5.4 \times 10^{-6}$	0.9510	0.9427	0.9582
Constant 100/2000	Bin4 CP	$0.26581 \pm 1.5 \times 10^{-4}$	0.9973	0.9947	0.9986
Constant 100/2000	Half CP	$0.24436 \pm 1.4 \times 10^{-4}$	0.9943	0.9909	0.9965
Constant 100/2000	Tails CP	$0.24436 \pm 1.4 \times 10^{-4}$	0.9943	0.9909	0.9965
Constant 100/2000	Label CP	$0.19348 \pm 1.1 \times 10^{-4}$	0.9547	0.9466	0.9615
Constant 100/2000	Label Hoeffding	$0.27162 \pm 1.0 \times 10^{-18}$	0.9973	0.9947	0.9986
Constant 100/2000	Label Wald	$0.18680 \pm 1.1 \times 10^{-4}$	0.9420	0.9331	0.9498
Constant 100/2000	Known-mass CP	$0.21921 \pm 1.2 \times 10^{-4}$	0.9810	0.9755	0.9853
Constant 100/2000	PPI	$0.18680 \pm 1.1 \times 10^{-4}$	0.9420	0.9331	0.9498
Constant 100/2000	PPI Hoeffding	$0.64800 \pm 3.7 \times 10^{-4}$	1.0000	0.9987	1.0000
Constant 100/2000	PPI++	$0.18680 \pm 1.1 \times 10^{-4}$	0.9420	0.9331	0.9498
Constant 100/2000	Safe CP	$0.24244 \pm 1.4 \times 10^{-4}$	0.9943	0.9909	0.9965
Constant 100/2000	Unsafe selection	$0.19348 \pm 1.1 \times 10^{-4}$	0.9547	0.9466	0.9615
Constant 100/2000	Bin8 Wald	$0.18680 \pm 1.1 \times 10^{-4}$	0.9420	0.9331	0.9498
Constant 100/2000	Bin4 Wald	$0.18680 \pm 1.1 \times 10^{-4}$	0.9420	0.9331	0.9498
Constant 100/2000	Half Wald	$0.18680 \pm 1.1 \times 10^{-4}$	0.9420	0.9331	0.9498
Constant 100/2000	Tails Wald	$0.18680 \pm 1.1 \times 10^{-4}$	0.9420	0.9331	0.9498
Constant 500/10000	Bin4 CP	$0.11852 \pm 3.0 \times 10^{-5}$	0.9967	0.9939	0.9982
Constant 500/10000	Half CP	$0.10849 \pm 2.7 \times 10^{-5}$	0.9920	0.9881	0.9946
Constant 500/10000	Tails CP	$0.10849 \pm 2.7 \times 10^{-5}$	0.9920	0.9881	0.9946
Constant 500/10000	Label CP	$0.08530 \pm 2.1 \times 10^{-5}$	0.9580	0.9502	0.9646
Constant 500/10000	Label Hoeffding	$0.12147 \pm 5.1 \times 10^{-19}$	0.9970	0.9943	0.9984
Constant 500/10000	Label Wald	$0.08359 \pm 2.2 \times 10^{-5}$	0.9520	0.9438	0.9591
Constant 500/10000	Known-mass CP	$0.09721 \pm 2.4 \times 10^{-5}$	0.9817	0.9762	0.9859
Constant 500/10000	PPI	$0.08359 \pm 2.2 \times 10^{-5}$	0.9520	0.9438	0.9591
Constant 500/10000	PPI Hoeffding	$0.29439 \pm 1.0 \times 10^{-18}$	1.0000	0.9987	1.0000
Constant 500/10000	PPI++	$0.08359 \pm 2.2 \times 10^{-5}$	0.9520	0.9438	0.9591
Constant 500/10000	Safe CP	$0.10803 \pm 2.7 \times 10^{-5}$	0.9903	0.9862	0.9933
Constant 500/10000	Unsafe selection	$0.08530 \pm 2.1 \times 10^{-5}$	0.9580	0.9502	0.9646
Constant 500/10000	Bin8 Wald	$0.08359 \pm 2.2 \times 10^{-5}$	0.9520	0.9438	0.9591
Constant 500/10000	Bin4 Wald	$0.08359 \pm 2.2 \times 10^{-5}$	0.9520	0.9438	0.9591
Constant 500/10000	Half Wald	$0.08359 \pm 2.2 \times 10^{-5}$	0.9520	0.9438	0.9591
Constant 500/10000	Tails Wald	$0.08359 \pm 2.2 \times 10^{-5}$	0.9520	0.9438	0.9591
Constant 2000/40000	Bin4 CP	$0.05890 \pm 7.5 \times 10^{-6}$	0.9937	0.9901	0.9959
Constant 2000/40000	Half CP	$0.05385 \pm 6.8 \times 10^{-6}$	0.9877	0.9830	0.9910
Constant 2000/40000	Tails CP	$0.05385 \pm 6.8 \times 10^{-6}$	0.9877	0.9830	0.9910
Constant 2000/40000	Label CP	$0.04228 \pm 5.3 \times 10^{-6}$	0.9623	0.9549	0.9686
Constant 2000/40000	Label Hoeffding	$0.06074 \pm 2.5 \times 10^{-19}$	0.9943	0.9909	0.9965
Constant 2000/40000	Label Wald	$0.04181 \pm 5.4 \times 10^{-6}$	0.9590	0.9513	0.9655
Constant 2000/40000	Known-mass CP	$0.04827 \pm 6.1 \times 10^{-6}$	0.9793	0.9736	0.9838
Constant 2000/40000	PPI	$0.04181 \pm 5.4 \times 10^{-6}$	0.9590	0.9513	0.9655
Constant 2000/40000	PPI Hoeffding	$0.14720 \pm 5.1 \times 10^{-19}$	1.0000	0.9987	1.0000
Constant 2000/40000	PPI++	$0.04181 \pm 5.4 \times 10^{-6}$	0.9590	0.9513	0.9655
Constant 2000/40000	Safe CP	$0.05373 \pm 6.8 \times 10^{-6}$	0.9877	0.9830	0.9910
Constant 2000/40000	Unsafe selection	$0.04228 \pm 5.3 \times 10^{-6}$	0.9623	0.9549	0.9686
Constant 2000/40000	Bin8 Wald	$0.04181 \pm 5.4 \times 10^{-6}$	0.9590	0.9513	0.9655
Constant 2000/40000	Bin4 Wald	$0.04181 \pm 5.4 \times 10^{-6}$	0.9590	0.9513	0.9655
Constant 2000/40000	Half Wald	$0.04181 \pm 5.4 \times 10^{-6}$	0.9590	0.9513	0.9655
Constant 2000/40000	Tails Wald	$0.04181 \pm 5.4 \times 10^{-6}$	0.9590	0.9513	0.9655
Rare tails 100/2000	Bin4 CP	$0.23127 \pm 3.4 \times 10^{-4}$	1.0000	0.9987	1.0000

Cell	Method	Width	Coverage	MC lower	MC upper
Rare tails 100/2000	Half CP	$0.11627 \pm 3.3 \times 10^{-4}$	0.9997	0.9981	0.9999
Rare tails 100/2000	Tails CP	$0.11581 \pm 3.1 \times 10^{-4}$	0.9980	0.9956	0.9991
Rare tails 100/2000	Label CP	$0.05208 \pm 2.6 \times 10^{-4}$	0.9820	0.9766	0.9862
Rare tails 100/2000	Label Hoeffding	$0.14565 \pm 1.8 \times 10^{-4}$	1.0000	0.9987	1.0000
Rare tails 100/2000	Label Wald	$0.02469 \pm 4.0 \times 10^{-4}$	0.6263	0.6089	0.6435
Rare tails 100/2000	Known-mass CP	$0.10116 \pm 2.8 \times 10^{-4}$	0.9957	0.9926	0.9975
Rare tails 100/2000	PPI	$0.06943 \pm 5.7 \times 10^{-4}$	0.9513	0.9430	0.9585
Rare tails 100/2000	PPI Hoeffding	$0.33803 \pm 5.7 \times 10^{-4}$	1.0000	0.9987	1.0000
Rare tails 100/2000	PPI++	$0.02449 \pm 4.0 \times 10^{-4}$	0.6260	0.6085	0.6431
Rare tails 100/2000	Safe CP	$0.06749 \pm 3.1 \times 10^{-4}$	0.9960	0.9930	0.9977
Rare tails 100/2000	Unsafe selection	$0.05208 \pm 2.6 \times 10^{-4}$	0.9820	0.9766	0.9862
Rare tails 100/2000	Bin8 Wald	$0.02398 \pm 3.9 \times 10^{-4}$	0.6217	0.6042	0.6389
Rare tails 100/2000	Bin4 Wald	$0.02436 \pm 3.9 \times 10^{-4}$	0.6240	0.6065	0.6412
Rare tails 100/2000	Half Wald	$0.02461 \pm 4.0 \times 10^{-4}$	0.6260	0.6085	0.6431
Rare tails 100/2000	Tails Wald	$0.02456 \pm 3.9 \times 10^{-4}$	0.6253	0.6079	0.6425
Rare tails 500/10000	Bin4 CP	$0.06384 \pm 1.3 \times 10^{-4}$	1.0000	0.9987	1.0000
Rare tails 500/10000	Half CP	$0.03710 \pm 1.2 \times 10^{-4}$	0.9997	0.9981	0.9999
Rare tails 500/10000	Tails CP	$0.03537 \pm 9.7 \times 10^{-5}$	0.9983	0.9961	0.9993
Rare tails 500/10000	Label CP	$0.01952 \pm 7.2 \times 10^{-5}$	0.9753	0.9691	0.9803
Rare tails 500/10000	Label Hoeffding	$0.07079 \pm 8.3 \times 10^{-5}$	1.0000	0.9987	1.0000
Rare tails 500/10000	Label Wald	$0.01665 \pm 8.6 \times 10^{-5}$	0.8647	0.8520	0.8764
Rare tails 500/10000	Known-mass CP	$0.03114 \pm 8.6 \times 10^{-5}$	0.9930	0.9893	0.9954
Rare tails 500/10000	PPI	$0.03677 \pm 2.4 \times 10^{-4}$	0.9450	0.9363	0.9526
Rare tails 500/10000	PPI Hoeffding	$0.15729 \pm 2.6 \times 10^{-4}$	1.0000	0.9987	1.0000
Rare tails 500/10000	PPI++	$0.01662 \pm 8.6 \times 10^{-5}$	0.8660	0.8533	0.8777
Rare tails 500/10000	Safe CP	$0.02463 \pm 8.9 \times 10^{-5}$	0.9947	0.9914	0.9967
Rare tails 500/10000	Unsafe selection	$0.01952 \pm 7.2 \times 10^{-5}$	0.9753	0.9691	0.9803
Rare tails 500/10000	Bin8 Wald	$0.01648 \pm 8.4 \times 10^{-5}$	0.8687	0.8561	0.8803
Rare tails 500/10000	Bin4 Wald	$0.01654 \pm 8.5 \times 10^{-5}$	0.8663	0.8537	0.8780
Rare tails 500/10000	Half Wald	$0.01664 \pm 8.6 \times 10^{-5}$	0.8647	0.8520	0.8764
Rare tails 500/10000	Tails Wald	$0.01658 \pm 8.5 \times 10^{-5}$	0.8657	0.8530	0.8774
Rare tails 2000/40000	Bin4 CP	$0.02499 \pm 4.0 \times 10^{-5}$	1.0000	0.9987	1.0000
Rare tails 2000/40000	Half CP	$0.01684 \pm 3.2 \times 10^{-5}$	1.0000	0.9987	1.0000
Rare tails 2000/40000	Tails CP	$0.01500 \pm 2.8 \times 10^{-5}$	0.9993	0.9976	0.9998
Rare tails 2000/40000	Label CP	$0.00920 \pm 1.8 \times 10^{-5}$	0.9527	0.9445	0.9597
Rare tails 2000/40000	Label Hoeffding	$0.04031 \pm 4.0 \times 10^{-5}$	1.0000	0.9987	1.0000
Rare tails 2000/40000	Label Wald	$0.00864 \pm 1.8 \times 10^{-5}$	0.9453	0.9366	0.9529
Rare tails 2000/40000	Known-mass CP	$0.01325 \pm 2.5 \times 10^{-5}$	0.9960	0.9930	0.9977
Rare tails 2000/40000	PPI	$0.02228 \pm 9.6 \times 10^{-5}$	0.9547	0.9466	0.9615
Rare tails 2000/40000	PPI Hoeffding	$0.08335 \pm 1.2 \times 10^{-4}$	1.0000	0.9987	1.0000
Rare tails 2000/40000	PPI++	$0.00864 \pm 1.8 \times 10^{-5}$	0.9370	0.9277	0.9451
Rare tails 2000/40000	Safe CP	$0.01163 \pm 2.2 \times 10^{-5}$	0.9927	0.9889	0.9952
Rare tails 2000/40000	Unsafe selection	$0.00920 \pm 1.8 \times 10^{-5}$	0.9527	0.9445	0.9597
Rare tails 2000/40000	Bin8 Wald	$0.00860 \pm 1.7 \times 10^{-5}$	0.9353	0.9260	0.9436
Rare tails 2000/40000	Bin4 Wald	$0.00860 \pm 1.8 \times 10^{-5}$	0.9353	0.9260	0.9436
Rare tails 2000/40000	Half Wald	$0.00864 \pm 1.8 \times 10^{-5}$	0.9410	0.9320	0.9489
Rare tails 2000/40000	Tails Wald	$0.00861 \pm 1.8 \times 10^{-5}$	0.9377	0.9284	0.9458
Shift 100/2000	Bin4 CP	$0.46289 \pm 8.3 \times 10^{-4}$	0.7167	0.7003	0.7325
Shift 100/2000	Half CP	$0.33252 \pm 4.3 \times 10^{-4}$	0.4340	0.4164	0.4518
Shift 100/2000	Tails CP	$0.36101 \pm 1.4 \times 10^{-4}$	0.9993	0.9976	0.9998
Shift 100/2000	Label CP	$0.20238 \pm 2.5 \times 10^{-5}$	0.9637	0.9564	0.9698
Shift 100/2000	Label Hoeffding	$0.27162 \pm 1.0 \times 10^{-18}$	0.9907	0.9865	0.9935
Shift 100/2000	Label Wald	$0.19597 \pm 2.6 \times 10^{-5}$	0.9460	0.9373	0.9535
Shift 100/2000	Known-mass CP	$0.32441 \pm 6.4 \times 10^{-5}$	0.9993	0.9976	0.9998
Shift 100/2000	PPI	$0.16840 \pm 1.8 \times 10^{-4}$	0.0000	0.0000	0.0013
Shift 100/2000	PPI Hoeffding	$0.52901 \pm 8.0 \times 10^{-4}$	0.7413	0.7254	0.7567
Shift 100/2000	PPI++	$0.03115 \pm 9.6 \times 10^{-4}$	0.0000	0.0000	0.0013
Shift 100/2000	Safe CP	$0.25364 \pm 3.2 \times 10^{-5}$	0.9850	0.9800	0.9888
Shift 100/2000	Unsafe selection	$0.20238 \pm 2.5 \times 10^{-5}$	0.9637	0.9564	0.9698

Cell	Method	Width	Coverage	MC lower	MC upper
Shift 100/2000	Bin8 Wald	$0.16266 \pm 1.7 \times 10^{-4}$	0.0030	0.0016	0.0057
Shift 100/2000	Bin4 Wald	$0.16741 \pm 1.6 \times 10^{-4}$	0.0050	0.0030	0.0082
Shift 100/2000	Half Wald	$0.17439 \pm 1.6 \times 10^{-4}$	0.0597	0.0517	0.0687
Shift 100/2000	Tails Wald	$0.19506 \pm 3.5 \times 10^{-5}$	0.9337	0.9242	0.9420
Shift 500/10000	Bin4 CP	$0.21271 \pm 1.9 \times 10^{-4}$	0.0000	0.0000	0.0013
Shift 500/10000	Half CP	$0.14937 \pm 8.9 \times 10^{-5}$	0.0000	0.0000	0.0013
Shift 500/10000	Tails CP	$0.16142 \pm 2.6 \times 10^{-5}$	0.9990	0.9971	0.9997
Shift 500/10000	Label CP	$0.08934 \pm 2.4 \times 10^{-6}$	0.9507	0.9423	0.9579
Shift 500/10000	Label Hoeffding	$0.12147 \pm 5.1 \times 10^{-19}$	0.9923	0.9885	0.9949
Shift 500/10000	Label Wald	$0.08765 \pm 2.4 \times 10^{-6}$	0.9420	0.9331	0.9498
Shift 500/10000	Known-mass CP	$0.14468 \pm 5.9 \times 10^{-6}$	0.9980	0.9956	0.9991
Shift 500/10000	PPI	$0.07529 \pm 3.6 \times 10^{-5}$	0.0000	0.0000	0.0013
Shift 500/10000	PPI Hoeffding	$0.29438 \pm 5.3 \times 10^{-6}$	0.0000	0.0000	0.0013
Shift 500/10000	PPI++	$0.00078 \pm 9.0 \times 10^{-5}$	0.0000	0.0000	0.0013
Shift 500/10000	Safe CP	$0.11317 \pm 3.0 \times 10^{-6}$	0.9893	0.9850	0.9924
Shift 500/10000	Unsafe selection	$0.08934 \pm 2.4 \times 10^{-6}$	0.9507	0.9423	0.9579
Shift 500/10000	Bin8 Wald	$0.07479 \pm 3.3 \times 10^{-5}$	0.0000	0.0000	0.0013
Shift 500/10000	Bin4 Wald	$0.07571 \pm 3.3 \times 10^{-5}$	0.0000	0.0000	0.0013
Shift 500/10000	Half Wald	$0.07833 \pm 3.3 \times 10^{-5}$	0.0000	0.0000	0.0013
Shift 500/10000	Tails Wald	$0.08757 \pm 3.2 \times 10^{-6}$	0.9273	0.9175	0.9361
Shift 2000/40000	Bin4 CP	$0.10591 \pm 4.7 \times 10^{-5}$	0.0000	0.0000	0.0013
Shift 2000/40000	Half CP	$0.07413 \pm 2.2 \times 10^{-5}$	0.0000	0.0000	0.0013
Shift 2000/40000	Tails CP	$0.08001 \pm 6.6 \times 10^{-6}$	0.9983	0.9961	0.9993
Shift 2000/40000	Label CP	$0.04429 \pm 2.8 \times 10^{-7}$	0.9530	0.9448	0.9600
Shift 2000/40000	Label Hoeffding	$0.06074 \pm 7.8 \times 10^{-19}$	0.9937	0.9901	0.9959
Shift 2000/40000	Label Wald	$0.04383 \pm 2.8 \times 10^{-7}$	0.9483	0.9398	0.9557
Shift 2000/40000	Known-mass CP	$0.07175 \pm 7.3 \times 10^{-7}$	0.9977	0.9952	0.9989
Shift 2000/40000	PPI	$0.03766 \pm 8.9 \times 10^{-6}$	0.0000	0.0000	0.0013
Shift 2000/40000	PPI Hoeffding	$0.14720 \pm 5.1 \times 10^{-19}$	0.0000	0.0000	0.0013
Shift 2000/40000	PPI++	0.00000 ± 0	0.0000	0.0000	0.0013
Shift 2000/40000	Safe CP	$0.05628 \pm 3.6 \times 10^{-7}$	0.9887	0.9842	0.9919
Shift 2000/40000	Unsafe selection	$0.04429 \pm 2.8 \times 10^{-7}$	0.9530	0.9448	0.9600
Shift 2000/40000	Bin8 Wald	$0.03760 \pm 8.1 \times 10^{-6}$	0.0000	0.0000	0.0013
Shift 2000/40000	Bin4 Wald	$0.03794 \pm 8.1 \times 10^{-6}$	0.0000	0.0000	0.0013
Shift 2000/40000	Half Wald	$0.03918 \pm 7.9 \times 10^{-6}$	0.0000	0.0000	0.0013
Shift 2000/40000	Tails Wald	$0.04382 \pm 3.9 \times 10^{-7}$	0.9347	0.9253	0.9430
Affine 500/500	Bin4 CP	$0.30333 \pm 1.1 \times 10^{-4}$	1.0000	0.9974	1.0000
Affine 500/500	Half CP	$0.19386 \pm 3.4 \times 10^{-5}$	1.0000	0.9974	1.0000
Affine 500/500	Tails CP	$0.16458 \pm 7.4 \times 10^{-5}$	0.9987	0.9952	0.9996
Affine 500/500	Label CP	$0.08934 \pm 3.5 \times 10^{-6}$	0.9520	0.9400	0.9617
Affine 500/500	Label Hoeffding	$0.12147 \pm 7.2 \times 10^{-19}$	0.9927	0.9869	0.9959
Affine 500/500	Label Wald	$0.08765 \pm 3.6 \times 10^{-6}$	0.9420	0.9290	0.9527
Affine 500/500	Known-mass CP	$0.14467 \pm 8.8 \times 10^{-6}$	0.9980	0.9941	0.9993
Affine 500/500	PPI	$0.09052 \pm 4.4 \times 10^{-5}$	0.9460	0.9334	0.9563
Affine 500/500	PPI Hoeffding	$0.39718 \pm 2.9 \times 10^{-18}$	1.0000	0.9974	1.0000
Affine 500/500	PPI++	$0.08149 \pm 2.2 \times 10^{-5}$	0.9493	0.9370	0.9593
Affine 500/500	Safe CP	$0.11316 \pm 4.5 \times 10^{-6}$	0.9900	0.9836	0.9939
Affine 500/500	Unsafe selection	$0.08934 \pm 3.5 \times 10^{-6}$	0.9520	0.9400	0.9617
Affine 500/500	Bin8 Wald	$0.08138 \pm 2.3 \times 10^{-5}$	0.9467	0.9341	0.9569
Affine 500/500	Bin4 Wald	$0.08181 \pm 2.2 \times 10^{-5}$	0.9487	0.9363	0.9587
Affine 500/500	Half Wald	$0.08304 \pm 2.2 \times 10^{-5}$	0.9513	0.9392	0.9611
Affine 500/500	Tails Wald	$0.08760 \pm 3.9 \times 10^{-6}$	0.9447	0.9319	0.9551
Affine 500/2500	Bin4 CP	$0.25404 \pm 8.5 \times 10^{-5}$	1.0000	0.9974	1.0000
Affine 500/2500	Half CP	$0.16608 \pm 3.1 \times 10^{-5}$	1.0000	0.9974	1.0000
Affine 500/2500	Tails CP	$0.16235 \pm 2.9 \times 10^{-5}$	1.0000	0.9974	1.0000
Affine 500/2500	Label CP	$0.08934 \pm 3.1 \times 10^{-6}$	0.9607	0.9496	0.9694
Affine 500/2500	Label Hoeffding	$0.12147 \pm 7.2 \times 10^{-19}$	0.9940	0.9886	0.9968
Affine 500/2500	Label Wald	$0.08765 \pm 3.1 \times 10^{-6}$	0.9460	0.9334	0.9563
Affine 500/2500	Known-mass CP	$0.14467 \pm 8.1 \times 10^{-6}$	1.0000	0.9974	1.0000

Cell	Method	Width	Coverage	MC lower	MC upper
Affine 500/2500	PPI	$0.07832 \pm 4.6 \times 10^{-5}$	0.9560	0.9444	0.9653
Affine 500/2500	PPI Hoeffding	$0.32400 \pm 1.4 \times 10^{-18}$	1.0000	0.9974	1.0000
Affine 500/2500	PPI++	$0.07708 \pm 3.6 \times 10^{-5}$	0.9593	0.9481	0.9682
Affine 500/2500	Safe CP	$0.11317 \pm 3.9 \times 10^{-6}$	0.9880	0.9811	0.9924
Affine 500/2500	Unsafe selection	$0.08934 \pm 3.1 \times 10^{-6}$	0.9607	0.9496	0.9694
Affine 500/2500	Bin8 Wald	$0.07687 \pm 3.7 \times 10^{-5}$	0.9580	0.9466	0.9670
Affine 500/2500	Bin4 Wald	$0.07764 \pm 3.7 \times 10^{-5}$	0.9567	0.9451	0.9659
Affine 500/2500	Half Wald	$0.07982 \pm 3.6 \times 10^{-5}$	0.9547	0.9429	0.9641
Affine 500/2500	Tails Wald	$0.08758 \pm 4.0 \times 10^{-6}$	0.9513	0.9392	0.9611
Affine 500/50000	Bin4 CP	$0.22278 \pm 1.2 \times 10^{-4}$	1.0000	0.9974	1.0000
Affine 500/50000	Half CP	$0.14888 \pm 7.1 \times 10^{-5}$	0.9993	0.9962	0.9999
Affine 500/50000	Tails CP	$0.16097 \pm 8.1 \times 10^{-6}$	1.0000	0.9974	1.0000
Affine 500/50000	Label CP	$0.08934 \pm 3.3 \times 10^{-6}$	0.9560	0.9444	0.9653
Affine 500/50000	Label Hoeffding	$0.12147 \pm 7.2 \times 10^{-19}$	0.9920	0.9861	0.9954
Affine 500/50000	Label Wald	$0.08765 \pm 3.3 \times 10^{-6}$	0.9453	0.9327	0.9557
Affine 500/50000	Known-mass CP	$0.14468 \pm 8.5 \times 10^{-6}$	0.9980	0.9941	0.9993
Affine 500/50000	PPI	$0.07524 \pm 5.0 \times 10^{-5}$	0.9573	0.9459	0.9664
Affine 500/50000	PPI Hoeffding	0.27803 ± 0	1.0000	0.9974	1.0000
Affine 500/50000	PPI++	$0.07500 \pm 4.5 \times 10^{-5}$	0.9547	0.9429	0.9641
Affine 500/50000	Safe CP	$0.11317 \pm 4.2 \times 10^{-6}$	0.9867	0.9795	0.9914
Affine 500/50000	Unsafe selection	$0.08934 \pm 3.3 \times 10^{-6}$	0.9560	0.9444	0.9653
Affine 500/50000	Bin8 Wald	$0.07472 \pm 4.5 \times 10^{-5}$	0.9500	0.9378	0.9599
Affine 500/50000	Bin4 Wald	$0.07566 \pm 4.5 \times 10^{-5}$	0.9527	0.9407	0.9623
Affine 500/50000	Half Wald	$0.07831 \pm 4.5 \times 10^{-5}$	0.9513	0.9392	0.9611
Affine 500/50000	Tails Wald	$0.08757 \pm 4.6 \times 10^{-6}$	0.9493	0.9370	0.9593
Sharp tails 500/500	Bin4 CP	$0.24795 \pm 1.2 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/500	Half CP	$0.16456 \pm 7.3 \times 10^{-5}$	1.0000	0.9974	1.0000
Sharp tails 500/500	Tails CP	$0.13919 \pm 1.2 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/500	Label CP	$0.08934 \pm 3.2 \times 10^{-6}$	0.9500	0.9378	0.9599
Sharp tails 500/500	Label Hoeffding	$0.12147 \pm 7.2 \times 10^{-19}$	0.9940	0.9886	0.9968
Sharp tails 500/500	Label Wald	$0.08765 \pm 3.3 \times 10^{-6}$	0.9373	0.9239	0.9485
Sharp tails 500/500	Known-mass CP	$0.02616 \pm 1.3 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/500	PPI	$0.11313 \pm 4.7 \times 10^{-5}$	0.9487	0.9363	0.9587
Sharp tails 500/500	PPI Hoeffding	$0.39718 \pm 2.9 \times 10^{-18}$	1.0000	0.9974	1.0000
Sharp tails 500/500	PPI++	$0.08761 \pm 3.6 \times 10^{-6}$	0.9407	0.9275	0.9515
Sharp tails 500/500	Safe CP	$0.11317 \pm 4.1 \times 10^{-6}$	0.9907	0.9844	0.9944
Sharp tails 500/500	Unsafe selection	$0.08934 \pm 3.2 \times 10^{-6}$	0.9500	0.9378	0.9599
Sharp tails 500/500	Bin8 Wald	$0.06259 \pm 1.0 \times 10^{-5}$	0.9513	0.9392	0.9611
Sharp tails 500/500	Bin4 Wald	$0.06259 \pm 1.0 \times 10^{-5}$	0.9520	0.9400	0.9617
Sharp tails 500/500	Half Wald	$0.08761 \pm 3.6 \times 10^{-6}$	0.9387	0.9254	0.9497
Sharp tails 500/500	Tails Wald	$0.06260 \pm 1.0 \times 10^{-5}$	0.9527	0.9407	0.9623
Sharp tails 500/2500	Bin4 CP	$0.14339 \pm 1.4 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/2500	Half CP	$0.16241 \pm 3.0 \times 10^{-5}$	0.9993	0.9962	0.9999
Sharp tails 500/2500	Tails CP	$0.07813 \pm 1.3 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/2500	Label CP	$0.08934 \pm 3.2 \times 10^{-6}$	0.9593	0.9481	0.9682
Sharp tails 500/2500	Label Hoeffding	$0.12147 \pm 7.2 \times 10^{-19}$	0.9920	0.9861	0.9954
Sharp tails 500/2500	Label Wald	$0.08766 \pm 3.2 \times 10^{-6}$	0.9480	0.9356	0.9581
Sharp tails 500/2500	Known-mass CP	$0.02607 \pm 1.3 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/2500	PPI	$0.10363 \pm 5.3 \times 10^{-5}$	0.9433	0.9305	0.9539
Sharp tails 500/2500	PPI Hoeffding	$0.32400 \pm 1.4 \times 10^{-18}$	1.0000	0.9974	1.0000
Sharp tails 500/2500	PPI++	$0.08758 \pm 4.4 \times 10^{-6}$	0.9507	0.9385	0.9605
Sharp tails 500/2500	Safe CP	$0.09291 \pm 1.4 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/2500	Unsafe selection	$0.07811 \pm 1.3 \times 10^{-4}$	0.9987	0.9952	0.9996
Sharp tails 500/2500	Bin8 Wald	$0.03748 \pm 2.7 \times 10^{-5}$	0.9560	0.9444	0.9653
Sharp tails 500/2500	Bin4 Wald	$0.03749 \pm 2.7 \times 10^{-5}$	0.9567	0.9451	0.9659
Sharp tails 500/2500	Half Wald	$0.08758 \pm 4.3 \times 10^{-6}$	0.9500	0.9378	0.9599
Sharp tails 500/2500	Tails Wald	$0.03750 \pm 2.7 \times 10^{-5}$	0.9567	0.9451	0.9659
Sharp tails 500/50000	Bin4 CP	$0.07537 \pm 1.5 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/50000	Half CP	$0.16096 \pm 7.9 \times 10^{-6}$	1.0000	0.9974	1.0000

Cell	Method	Width	Coverage	MC lower	MC upper
Sharp tails 500/50000	Tails CP	$0.04031 \pm 1.3 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/50000	Label CP	$0.08934 \pm 3.2 \times 10^{-6}$	0.9560	0.9444	0.9653
Sharp tails 500/50000	Label Hoeffding	$0.12147 \pm 7.2 \times 10^{-19}$	0.9933	0.9878	0.9964
Sharp tails 500/50000	Label Wald	$0.08765 \pm 3.2 \times 10^{-6}$	0.9453	0.9327	0.9557
Sharp tails 500/50000	Known-mass CP	$0.02619 \pm 1.3 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/50000	PPI	$0.10139 \pm 5.4 \times 10^{-5}$	0.9473	0.9348	0.9575
Sharp tails 500/50000	PPI Hoeffding	0.27803 ± 0	1.0000	0.9974	1.0000
Sharp tails 500/50000	PPI++	$0.08756 \pm 4.7 \times 10^{-6}$	0.9487	0.9363	0.9587
Sharp tails 500/50000	Safe CP	$0.04852 \pm 1.5 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/50000	Unsafe selection	$0.04031 \pm 1.3 \times 10^{-4}$	1.0000	0.9974	1.0000
Sharp tails 500/50000	Bin8 Wald	$0.01475 \pm 8.0 \times 10^{-5}$	0.9467	0.9341	0.9569
Sharp tails 500/50000	Bin4 Wald	$0.01479 \pm 8.1 \times 10^{-5}$	0.9507	0.9385	0.9605
Sharp tails 500/50000	Half Wald	$0.08756 \pm 4.7 \times 10^{-6}$	0.9493	0.9370	0.9593
Sharp tails 500/50000	Tails Wald	$0.01481 \pm 8.1 \times 10^{-5}$	0.9527	0.9407	0.9623
Irrelevant 500/500	Bin4 CP	$0.24824 \pm 1.4 \times 10^{-4}$	1.0000	0.9974	1.0000
Irrelevant 500/500	Half CP	$0.15716 \pm 8.9 \times 10^{-5}$	1.0000	0.9974	1.0000
Irrelevant 500/500	Tails CP	$0.15704 \pm 8.5 \times 10^{-5}$	1.0000	0.9974	1.0000
Irrelevant 500/500	Label CP	$0.08528 \pm 3.0 \times 10^{-5}$	0.9627	0.9518	0.9711
Irrelevant 500/500	Label Hoeffding	$0.12147 \pm 7.2 \times 10^{-19}$	0.9960	0.9913	0.9982
Irrelevant 500/500	Label Wald	$0.08357 \pm 3.1 \times 10^{-5}$	0.9560	0.9444	0.9653
Irrelevant 500/500	Known-mass CP	$0.13815 \pm 5.0 \times 10^{-5}$	0.9987	0.9952	0.9996
Irrelevant 500/500	PPI	$0.11001 \pm 5.4 \times 10^{-5}$	0.9567	0.9451	0.9659
Irrelevant 500/500	PPI Hoeffding	$0.39718 \pm 2.9 \times 10^{-18}$	1.0000	0.9974	1.0000
Irrelevant 500/500	PPI++	$0.08353 \pm 3.1 \times 10^{-5}$	0.9547	0.9429	0.9641
Irrelevant 500/500	Safe CP	$0.10801 \pm 3.9 \times 10^{-5}$	0.9893	0.9827	0.9934
Irrelevant 500/500	Unsafe selection	$0.08528 \pm 3.0 \times 10^{-5}$	0.9627	0.9518	0.9711
Irrelevant 500/500	Bin8 Wald	$0.08327 \pm 3.1 \times 10^{-5}$	0.9480	0.9356	0.9581
Irrelevant 500/500	Bin4 Wald	$0.08344 \pm 3.1 \times 10^{-5}$	0.9500	0.9378	0.9599
Irrelevant 500/500	Half Wald	$0.08352 \pm 3.1 \times 10^{-5}$	0.9540	0.9422	0.9635
Irrelevant 500/500	Tails Wald	$0.08353 \pm 3.1 \times 10^{-5}$	0.9547	0.9429	0.9641
Irrelevant 500/2500	Bin4 CP	$0.24195 \pm 9.5 \times 10^{-5}$	1.0000	0.9974	1.0000
Irrelevant 500/2500	Half CP	$0.15515 \pm 6.1 \times 10^{-5}$	1.0000	0.9974	1.0000
Irrelevant 500/2500	Tails CP	$0.15508 \pm 6.0 \times 10^{-5}$	1.0000	0.9974	1.0000
Irrelevant 500/2500	Label CP	$0.08535 \pm 3.0 \times 10^{-5}$	0.9613	0.9503	0.9700
Irrelevant 500/2500	Label Hoeffding	$0.12147 \pm 7.2 \times 10^{-19}$	0.9987	0.9952	0.9996
Irrelevant 500/2500	Label Wald	$0.08364 \pm 3.0 \times 10^{-5}$	0.9547	0.9429	0.9641
Irrelevant 500/2500	Known-mass CP	$0.13827 \pm 4.9 \times 10^{-5}$	1.0000	0.9974	1.0000
Irrelevant 500/2500	PPI	$0.10029 \pm 5.6 \times 10^{-5}$	0.9553	0.9437	0.9647
Irrelevant 500/2500	PPI Hoeffding	$0.32400 \pm 1.4 \times 10^{-18}$	1.0000	0.9974	1.0000
Irrelevant 500/2500	PPI++	$0.08357 \pm 3.0 \times 10^{-5}$	0.9513	0.9392	0.9611
Irrelevant 500/2500	Safe CP	$0.10810 \pm 3.8 \times 10^{-5}$	0.9913	0.9852	0.9949
Irrelevant 500/2500	Unsafe selection	$0.08535 \pm 3.0 \times 10^{-5}$	0.9613	0.9503	0.9700
Irrelevant 500/2500	Bin8 Wald	$0.08315 \pm 3.1 \times 10^{-5}$	0.9533	0.9415	0.9629
Irrelevant 500/2500	Bin4 Wald	$0.08344 \pm 3.1 \times 10^{-5}$	0.9540	0.9422	0.9635
Irrelevant 500/2500	Half Wald	$0.08357 \pm 3.0 \times 10^{-5}$	0.9520	0.9400	0.9617
Irrelevant 500/2500	Tails Wald	$0.08357 \pm 3.0 \times 10^{-5}$	0.9513	0.9392	0.9611
Irrelevant 500/50000	Bin4 CP	$0.23778 \pm 8.4 \times 10^{-5}$	1.0000	0.9974	1.0000
Irrelevant 500/50000	Half CP	$0.15367 \pm 5.4 \times 10^{-5}$	1.0000	0.9974	1.0000
Irrelevant 500/50000	Tails CP	$0.15365 \pm 5.4 \times 10^{-5}$	1.0000	0.9974	1.0000
Irrelevant 500/50000	Label CP	$0.08527 \pm 3.0 \times 10^{-5}$	0.9593	0.9481	0.9682
Irrelevant 500/50000	Label Hoeffding	$0.12147 \pm 7.2 \times 10^{-19}$	0.9967	0.9922	0.9986
Irrelevant 500/50000	Label Wald	$0.08356 \pm 3.0 \times 10^{-5}$	0.9553	0.9437	0.9647
Irrelevant 500/50000	Known-mass CP	$0.13811 \pm 4.9 \times 10^{-5}$	1.0000	0.9974	1.0000
Irrelevant 500/50000	PPI	$0.09779 \pm 6.0 \times 10^{-5}$	0.9573	0.9459	0.9664
Irrelevant 500/50000	PPI Hoeffding	0.27803 ± 0	1.0000	0.9974	1.0000
Irrelevant 500/50000	PPI++	$0.08347 \pm 3.0 \times 10^{-5}$	0.9527	0.9407	0.9623
Irrelevant 500/50000	Safe CP	$0.10799 \pm 3.8 \times 10^{-5}$	0.9920	0.9861	0.9954
Irrelevant 500/50000	Unsafe selection	$0.08527 \pm 3.0 \times 10^{-5}$	0.9593	0.9481	0.9682
Irrelevant 500/50000	Bin8 Wald	$0.08298 \pm 3.1 \times 10^{-5}$	0.9513	0.9392	0.9611

Cell	Method	Width	Coverage	MC lower	MC upper
Irrelevant 500/50000	Bin4 Wald	$0.08331 \pm 3.1 \times 10^{-5}$	0.9533	0.9415	0.9629
Irrelevant 500/50000	Half Wald	$0.08347 \pm 3.0 \times 10^{-5}$	0.9540	0.9422	0.9635
Irrelevant 500/50000	Tails Wald	$0.08347 \pm 3.0 \times 10^{-5}$	0.9553	0.9437	0.9647
Forest 100/2000	Bin4 CP	$0.29922 \pm 4.3 \times 10^{-4}$	1.0000	0.9987	1.0000
Forest 100/2000	Half CP	$0.22491 \pm 3.7 \times 10^{-4}$	1.0000	0.9987	1.0000
Forest 100/2000	Tails CP	$0.24884 \pm 3.8 \times 10^{-4}$	1.0000	0.9987	1.0000
Forest 100/2000	Label CP	$0.14851 \pm 2.5 \times 10^{-4}$	0.9443	0.9355	0.9520
Forest 100/2000	Label Hoeffding	$0.26438 \pm 2.6 \times 10^{-4}$	0.9997	0.9981	0.9999
Forest 100/2000	Label Wald	$0.14030 \pm 2.6 \times 10^{-4}$	0.9350	0.9256	0.9433
Forest 100/2000	Known-mass CP	$0.21838 \pm 3.3 \times 10^{-4}$	0.9993	0.9976	0.9998
Forest 100/2000	PPI	$0.11974 \pm 2.6 \times 10^{-4}$	0.9400	0.9309	0.9479
Forest 100/2000	PPI Hoeffding	$0.48172 \pm 5.6 \times 10^{-4}$	1.0000	0.9987	1.0000
Forest 100/2000	PPI++	$0.12003 \pm 2.6 \times 10^{-4}$	0.9227	0.9126	0.9317
Forest 100/2000	Safe CP	$0.18585 \pm 3.1 \times 10^{-4}$	0.9897	0.9854	0.9927
Forest 100/2000	Unsafe selection	$0.14850 \pm 2.5 \times 10^{-4}$	0.9443	0.9355	0.9520
Forest 100/2000	Bin8 Wald	$0.11200 \pm 2.7 \times 10^{-4}$	0.8950	0.8835	0.9055
Forest 100/2000	Bin4 Wald	$0.11795 \pm 2.8 \times 10^{-4}$	0.9203	0.9101	0.9295
Forest 100/2000	Half Wald	$0.12375 \pm 2.8 \times 10^{-4}$	0.9310	0.9214	0.9395
Forest 100/2000	Tails Wald	$0.13734 \pm 2.5 \times 10^{-4}$	0.9327	0.9231	0.9411
Forest 500/10000	Bin4 CP	$0.14650 \pm 1.2 \times 10^{-4}$	1.0000	0.9987	1.0000
Forest 500/10000	Half CP	$0.10352 \pm 8.2 \times 10^{-5}$	0.9997	0.9981	0.9999
Forest 500/10000	Tails CP	$0.11167 \pm 8.5 \times 10^{-5}$	1.0000	0.9987	1.0000
Forest 500/10000	Label CP	$0.06461 \pm 4.9 \times 10^{-5}$	0.9607	0.9531	0.9671
Forest 500/10000	Label Hoeffding	$0.12147 \pm 5.1 \times 10^{-19}$	1.0000	0.9987	1.0000
Forest 500/10000	Label Wald	$0.06276 \pm 4.9 \times 10^{-5}$	0.9480	0.9395	0.9554
Forest 500/10000	Known-mass CP	$0.09790 \pm 7.2 \times 10^{-5}$	0.9993	0.9976	0.9998
Forest 500/10000	PPI	$0.05369 \pm 5.3 \times 10^{-5}$	0.9487	0.9402	0.9560
Forest 500/10000	PPI Hoeffding	$0.29073 \pm 1.2 \times 10^{-4}$	1.0000	0.9987	1.0000
Forest 500/10000	PPI++	$0.05346 \pm 5.1 \times 10^{-5}$	0.9450	0.9363	0.9526
Forest 500/10000	Safe CP	$0.08173 \pm 6.2 \times 10^{-5}$	0.9927	0.9889	0.9952
Forest 500/10000	Unsafe selection	$0.06461 \pm 4.9 \times 10^{-5}$	0.9607	0.9531	0.9671
Forest 500/10000	Bin8 Wald	$0.05267 \pm 5.2 \times 10^{-5}$	0.9407	0.9316	0.9486
Forest 500/10000	Bin4 Wald	$0.05405 \pm 5.4 \times 10^{-5}$	0.9443	0.9355	0.9520
Forest 500/10000	Half Wald	$0.05589 \pm 5.4 \times 10^{-5}$	0.9453	0.9366	0.9529
Forest 500/10000	Tails Wald	$0.06182 \pm 4.9 \times 10^{-5}$	0.9520	0.9438	0.9591
Forest 2000/40000	Bin4 CP	$0.07409 \pm 3.2 \times 10^{-5}$	1.0000	0.9987	1.0000
Forest 2000/40000	Half CP	$0.05159 \pm 2.0 \times 10^{-5}$	0.9997	0.9981	0.9999
Forest 2000/40000	Tails CP	$0.05538 \pm 2.2 \times 10^{-5}$	0.9987	0.9966	0.9995
Forest 2000/40000	Label CP	$0.03189 \pm 1.3 \times 10^{-5}$	0.9540	0.9459	0.9609
Forest 2000/40000	Label Hoeffding	$0.06074 \pm 2.5 \times 10^{-19}$	1.0000	0.9987	1.0000
Forest 2000/40000	Label Wald	$0.03141 \pm 1.3 \times 10^{-5}$	0.9483	0.9398	0.9557
Forest 2000/40000	Known-mass CP	$0.04845 \pm 1.9 \times 10^{-5}$	0.9953	0.9922	0.9972
Forest 2000/40000	PPI	$0.02690 \pm 1.3 \times 10^{-5}$	0.9507	0.9423	0.9579
Forest 2000/40000	PPI Hoeffding	$0.14720 \pm 5.1 \times 10^{-19}$	1.0000	0.9987	1.0000
Forest 2000/40000	PPI++	$0.02676 \pm 1.2 \times 10^{-5}$	0.9500	0.9416	0.9572
Forest 2000/40000	Safe CP	$0.04050 \pm 1.6 \times 10^{-5}$	0.9837	0.9785	0.9876
Forest 2000/40000	Unsafe selection	$0.03189 \pm 1.3 \times 10^{-5}$	0.9540	0.9459	0.9609
Forest 2000/40000	Bin8 Wald	$0.02665 \pm 1.3 \times 10^{-5}$	0.9500	0.9416	0.9572
Forest 2000/40000	Bin4 Wald	$0.02718 \pm 1.3 \times 10^{-5}$	0.9547	0.9466	0.9615
Forest 2000/40000	Half Wald	$0.02803 \pm 1.3 \times 10^{-5}$	0.9493	0.9409	0.9566
Forest 2000/40000	Tails Wald	$0.03098 \pm 1.2 \times 10^{-5}$	0.9503	0.9420	0.9575
Galaxies 100/2000	Bin4 CP	$0.35245 \pm 4.5 \times 10^{-4}$	1.0000	0.9987	1.0000
Galaxies 100/2000	Half CP	$0.26093 \pm 3.9 \times 10^{-4}$	1.0000	0.9987	1.0000
Galaxies 100/2000	Tails CP	$0.30337 \pm 2.8 \times 10^{-4}$	1.0000	0.9987	1.0000
Galaxies 100/2000	Label CP	$0.17851 \pm 1.7 \times 10^{-4}$	0.9533	0.9452	0.9603
Galaxies 100/2000	Label Hoeffding	$0.27161 \pm 6.2 \times 10^{-6}$	0.9977	0.9952	0.9989
Galaxies 100/2000	Label Wald	$0.17135 \pm 1.8 \times 10^{-4}$	0.9373	0.9281	0.9455
Galaxies 100/2000	Known-mass CP	$0.26679 \pm 2.6 \times 10^{-4}$	0.9993	0.9976	0.9998
Galaxies 100/2000	PPI	$0.11832 \pm 2.4 \times 10^{-4}$	0.9463	0.9377	0.9538

Cell	Method	Width	Coverage	MC lower	MC upper
Galaxies 100/2000	PPI Hoeffding	$0.58826 \pm 5.5 \times 10^{-4}$	1.0000	0.9987	1.0000
Galaxies 100/2000	PPI++	$0.11870 \pm 2.2 \times 10^{-4}$	0.9337	0.9242	0.9420
Galaxies 100/2000	Safe CP	$0.22357 \pm 2.2 \times 10^{-4}$	0.9923	0.9885	0.9949
Galaxies 100/2000	Unsafe selection	$0.17851 \pm 1.7 \times 10^{-4}$	0.9533	0.9452	0.9603
Galaxies 100/2000	Bin8 Wald	$0.11094 \pm 2.5 \times 10^{-4}$	0.8930	0.8814	0.9036
Galaxies 100/2000	Bin4 Wald	$0.11723 \pm 2.5 \times 10^{-4}$	0.9280	0.9182	0.9367
Galaxies 100/2000	Half Wald	$0.12708 \pm 2.6 \times 10^{-4}$	0.9417	0.9327	0.9495
Galaxies 100/2000	Tails Wald	$0.16701 \pm 1.8 \times 10^{-4}$	0.9347	0.9253	0.9430
Galaxies 500/10000	Bin4 CP	$0.16787 \pm 1.2 \times 10^{-4}$	1.0000	0.9987	1.0000
Galaxies 500/10000	Half CP	$0.11715 \pm 8.2 \times 10^{-5}$	1.0000	0.9987	1.0000
Galaxies 500/10000	Tails CP	$0.13716 \pm 6.1 \times 10^{-5}$	0.9997	0.9981	0.9999
Galaxies 500/10000	Label CP	$0.07853 \pm 3.5 \times 10^{-5}$	0.9490	0.9405	0.9563
Galaxies 500/10000	Label Hoeffding	$0.12147 \pm 5.1 \times 10^{-19}$	0.9987	0.9966	0.9995
Galaxies 500/10000	Label Wald	$0.07677 \pm 3.6 \times 10^{-5}$	0.9407	0.9316	0.9486
Galaxies 500/10000	Known-mass CP	$0.11973 \pm 5.4 \times 10^{-5}$	0.9987	0.9966	0.9995
Galaxies 500/10000	PPI	$0.05318 \pm 4.6 \times 10^{-5}$	0.9477	0.9391	0.9551
Galaxies 500/10000	PPI Hoeffding	$0.29439 \pm 1.0 \times 10^{-18}$	1.0000	0.9987	1.0000
Galaxies 500/10000	PPI++	$0.05309 \pm 4.4 \times 10^{-5}$	0.9440	0.9352	0.9517
Galaxies 500/10000	Safe CP	$0.09942 \pm 4.5 \times 10^{-5}$	0.9907	0.9865	0.9935
Galaxies 500/10000	Unsafe selection	$0.07853 \pm 3.5 \times 10^{-5}$	0.9490	0.9405	0.9563
Galaxies 500/10000	Bin8 Wald	$0.05265 \pm 4.6 \times 10^{-5}$	0.9390	0.9299	0.9470
Galaxies 500/10000	Bin4 Wald	$0.05388 \pm 4.8 \times 10^{-5}$	0.9440	0.9352	0.9517
Galaxies 500/10000	Half Wald	$0.05742 \pm 5.1 \times 10^{-5}$	0.9487	0.9402	0.9560
Galaxies 500/10000	Tails Wald	$0.07520 \pm 3.6 \times 10^{-5}$	0.9430	0.9341	0.9507
Galaxies 2000/40000	Bin4 CP	$0.08399 \pm 3.1 \times 10^{-5}$	1.0000	0.9987	1.0000
Galaxies 2000/40000	Half CP	$0.05802 \pm 2.1 \times 10^{-5}$	1.0000	0.9987	1.0000
Galaxies 2000/40000	Tails CP	$0.06818 \pm 1.5 \times 10^{-5}$	0.9997	0.9981	0.9999
Galaxies 2000/40000	Label CP	$0.03889 \pm 8.5 \times 10^{-6}$	0.9573	0.9495	0.9640
Galaxies 2000/40000	Label Hoeffding	$0.06074 \pm 3.6 \times 10^{-19}$	0.9993	0.9976	0.9998
Galaxies 2000/40000	Label Wald	$0.03842 \pm 8.5 \times 10^{-6}$	0.9523	0.9441	0.9594
Galaxies 2000/40000	Known-mass CP	$0.05935 \pm 1.3 \times 10^{-5}$	0.9993	0.9976	0.9998
Galaxies 2000/40000	PPI	$0.02660 \pm 1.1 \times 10^{-5}$	0.9437	0.9348	0.9514
Galaxies 2000/40000	PPI Hoeffding	$0.14720 \pm 5.1 \times 10^{-19}$	1.0000	0.9987	1.0000
Galaxies 2000/40000	PPI++	$0.02653 \pm 1.1 \times 10^{-5}$	0.9427	0.9338	0.9504
Galaxies 2000/40000	Safe CP	$0.04941 \pm 1.1 \times 10^{-5}$	0.9927	0.9889	0.9952
Galaxies 2000/40000	Unsafe selection	$0.03889 \pm 8.5 \times 10^{-6}$	0.9573	0.9495	0.9640
Galaxies 2000/40000	Bin8 Wald	$0.02660 \pm 1.1 \times 10^{-5}$	0.9427	0.9338	0.9504
Galaxies 2000/40000	Bin4 Wald	$0.02706 \pm 1.2 \times 10^{-5}$	0.9477	0.9391	0.9551
Galaxies 2000/40000	Half Wald	$0.02874 \pm 1.3 \times 10^{-5}$	0.9457	0.9370	0.9532
Galaxies 2000/40000	Tails Wald	$0.03767 \pm 8.8 \times 10^{-6}$	0.9550	0.9470	0.9619

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